When Innovation Outpaces Policy: Hyperscale Data Centers and the Local Costs of Investment Incentives*

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Abstract

Governments increasingly use tax incentives to attract capital-intensive projects, yet their local returns remain uncertain. I study how these projects affect public finances, employment, local resources, and political outcomes—and whether these effects change as technological innovation increases project scale and resource demands. Larger projects may expand the local tax base and generate complementary economic activity, but their capital intensity may limit employment gains, while tax expenditures, infrastructure needs, and competition for electricity, water, and land may increase with project scale. Exploiting the staggered adoption of state data center tax incentives and county-level variation in proximity to major water bodies, I find that subsidies increase local government revenues, expenditures, and debt issuance. Overall employment declines, even though employment rises in sectors directly exposed to data center activity. Combined with the observed changes in local resource prices, this pattern is consistent with data centers competing for scarce resources and crowding out employment in other local industries. Among states adopting subsidies after the transition toward larger facilities, exposure raises developed-land prices and residential electricity rates while lowering farmland prices, consistent with growing concerns about water scarcity. These findings show that incentives for capital-intensive investment can generate local fiscal benefits while imposing employment, infrastructure, resource, and political costs that intensify—and, in some cases, reverse—as technological change increases project scale.

Keywords: Data Centers; Public Finance; Employment; Local Politics; Energy Demand; Land Price; Regional Economic Activity

JEL Codes: G38; H25; H71; H72; R3; R11; R38; O33; Q41

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1 Introduction

The rapid expansion of capital-intensive technologies and recent policy proposals[1] have renewed debate over using tax incentives to encourage capital investment. These incentives have become increasingly prominent (Acemoglu et al., 2020) and contentious [2]. These provisions condition tax benefits primarily on the purchase and ownership of capital, rather than on specific local outcomes such as job creation (Ku et al., 2020; Neumark and Grijalva, 2017), worker training (Abramovsky et al., 2011; Martins, 2021), or environmental remediation (Swenson, 2019). Tax relief can therefore increase with the scale of a project even when employment and other local benefits do not, creating scope for unintended consequences. Despite their widespread adoption, whether these incentives generate meaningful local economic benefits remains an open empirical question.

Policymakers generally justify these incentives as a means of increasing local employment and expanding the tax base.[3] However, whether these incentives achieve either objective is not obvious ex ante. First, their benefits depend on whether they induce investment that would not otherwise occur (Hall and Jorgenson, 1967; Cummins et al., 1994; House and Shapiro, 2008; Edgerton, 2010; Zwick and Mahon, 2017; Maffini et al., 2019; Ohrn, 2019). When a project would have proceeded without the incentive, the government forgoes revenue without altering the firm's investment decision.

Second, a larger tax base may strengthen local fiscal capacity, support spending on amenities for residents, and lower municipal borrowing costs by improving perceived credit quality (Schwert, 2017; Gao et al., 2020). These fiscal benefits, however, are not guaranteed. Tax exemptions limit the revenue collected from new capital, while subsidized projects may require additional spending on infrastructure and public services. Governments may finance these costs by issuing debt, potentially straining debt capacity and raising borrowing costs (Chava et al., 2024), or by increasing taxes and cutting spending elsewhere (Bartik, 2019). A larger tax base therefore need not translate into a stronger fiscal position.

Third, even when an incentive increases investment, its employment effects depend on whether the new capital complements or substitutes for local labor (Garrett et al., 2020). Finally, subsidized projects may generate agglomeration spillovers (Dougal et al., 2015) and broader local multiplier effects (Moretti, 2010). At the same time, they may compete with incumbent firms and residents for scarce resources, raise local costs, and displace existing economic activity. The consequences for host communities are therefore uncertain even when these incentives succeed in attracting additional investment.

Technological change complicates these predictions because investment incentives may remain unchanged as the scale and resource requirements of the projects they attract evolve. Larger projects can generate greater tax revenue and complementary economic activity, but these benefits may not increase proportionately with investment. Employment may grow more slowly than capital, while demands on electricity, water, land, infrastructure, and public services may increase substantially. The local effects of an investment incentive may therefore change in magnitude or direction as projects become larger and more resource-intensive. This creates a particular challenge when incentives are designed for emerging industries whose future development is uncertain. Policymakers may have limited ability to anticipate how rapidly technology will evolve, how large future projects will become, or how many investments a policy will subsidize. If statutory provisions remain unchanged, the same incentives may apply to successive generations of projects that are substantially larger, replace equipment more frequently, and consume more local resources than originally anticipated. Policies designed to attract an emerging industry may therefore generate

[1]For example, a 2025 U.S. executive order directed the Secretary of Commerce to develop a program providing financial support—including loans, grants, and tax incentives—for data centers and related infrastructure projects.

[2]Examples of statutory investment incentives include accelerated depreciation under Internal Revenue Code (I.R.C.) § 168(k) and elective expensing of qualifying depreciable property under I.R.C. § 179. Sections 13201 and 13101 of the Tax Cuts and Jobs Act of 2017 expanded these provisions, respectively. The CHIPS Act of 2022 created an investment tax credit under I.R.C. § 48D for qualified property used in semiconductor manufacturing facilities.

[3]Washington's 2022 data center sales- and use-tax exemption provides a clear example. The legislature stated that the policy was intended to encourage investment, thereby "growing employment in the technology industry while adding real and personal property to state and local property tax rolls, thereby increasing the county tax base." 2022 Wash. Sess. Laws ch. 267, §§ 1–2; see also Wash. Rev. Code §§ 82.08.986–82.08.9861 and 82.12.986–82.12.9861.

unintended local consequences as that industry develops.

This paper examines these effects by asking two broad questions about capital-intensive development policy. First, how do large capital-intensive projects affect local economies, governments, and residents? Second, how do these consequences change as technological innovation increases the scale and resource intensity of subsidized investment? These questions are central to evaluating incentives targeting rapidly evolving, capital-intensive industries.

I study these questions through the expansion of data centers in the United States, using the industry as a lens on capital-intensive development more broadly. Data centers require substantial upfront and recurring capital expenditures, and policymakers often view them as opportunities to expand the local tax base, stimulate construction activity, and position their regions within the growing digital and artificial-intelligence economy. In response, many states have adopted targeted incentives that reduce the cost of data center investment through sales- and property-tax exemptions for equipment, facilities, electricity, and other qualifying inputs (Table B.2). As of 2025, 40 states offer some form of data center incentive (Figure 1), making data centers one of the most heavily subsidized categories of modern infrastructure.

Evaluating investment tax incentives presents several empirical challenges. First, many investment policies are adopted nationally, leaving little geographic variation with which to construct a comparison group. Second, these incentives are often enacted as part of broader legislative packages, making it difficult to isolate their effects from those of other policy changes introduced at the same time.[4] To address these challenges, I exploit the staggered adoption of data center incentives across states. For each adoption year, I construct a separate comparison sample containing states that adopt in that year and states that have not yet adopted or never adopt an incentive. I then combine these samples in a stacked

[4]For example, the Tax Cuts and Jobs Act of 2017 expanded investment expensing alongside reductions in corporate income tax rates, new limits on interest deductions, and changes to the taxation of foreign earnings. Similarly, the CHIPS and Science Act of 2022 introduced a semiconductor investment tax credit alongside direct manufacturing subsidies and funding for research and development and workforce training.

design (Gormley and Matsa, 2011).

This approach, however, leaves open the possibility that states adopting these policies differ from other states or have concurrent policies changes. To mitigate this concern, I use within-state variation in counties' exposure to the policy, comparing water-adjacent versus non-water-adjacent counties counties subject to the same state-level policy environment. I classify counties that contain or border a coast, lake, or river as water-adjacent counties. I use water-adjacent status as a predetermined measure of suitability because access to water is important for cooling large data center facilities.

The specification includes three sets of fixed effects, each addressing a different concern. County-by-cohort fixed effects account for time-invariant county characteristics, such as physical geography and proximity to major population centers. State-by-year-by-cohort fixed effects account for contemporaneous statewide changes, such as a broader corporate tax reform, a statewide economic downturn, or a change in political leadership occurring in the year an incentive is adopted. Water-by-year-by-cohort fixed effects account for changes that affect water-adjacent counties generally, such as changes in federal environmental regulations, flood-insurance policy, or water-adjacent land values. After accounting for these factors, the estimate compares the change in outcomes between water-adjacent versus non-water-adjacent counties in adopting states with the corresponding change in control states.The identifying parallel trends assumption is that, absent the adoption of data center tax incentives, the relative outcomes of water-adjacent versus non-water-adjacent counties would have evolved similarly across treated and control states.

To mitigate concerns that adopting and non-adopting states differ systematically in their economic characteristics or trends, I assess their comparability in two ways. First, I compare the distributions of observable economic characteristics across states in my sample and find that adopting and non-adopting states are broadly similar (Figure A.1). Second, I compare state-level economic trends for eventually treated and never-treated states over the sample period and find similar patterns across the two groups (Figure A.2). These

comparisons provide further support for the comparability of adopting and non-adopting states.

Another potential concern is that water-adjacent counties may differ systematically from non-water-adjacent counties in characteristics that independently affect local economic outcomes.To address this concern, I compare the two groups using demographic and economic characteristics measured in year 2000, the first year of treatment. I find no statistically significant differences in minority share, female share, poverty, unemployment, information-sector employment, median age, or average commute time (Table B.3).

I begin by examining whether the empirical design captures increases in data center development activity. I use novel data from Aterio, supplemented with hand-collected information from Baxtel.com to fill in missing facilities and facility details, including announcement dates, estimated space, and power capacity. The estimated increase in data center count equals 34.1 percent of its sample mean.[5] The corresponding increases in facility space and power capacity equal 36.5 and 70.2 percent of their respective sample means. These results suggest that incentives attract more data center projects and expand their overall capacity.[6]

To answer the paper's first question of how large, capital intensive projects affect local economies, governments, and residents, I examine three margins: local public finances, employment, and resource costs.

Following incentive adoption, waterfront counties in adopting states experience a 2.0 percent increase in total revenue and a 2.5 percent increase in total taxes relative to the comparison counties. I do not find statistically significant changes in general charges or intergovernmental revenue. This pattern points to increased tax collections rather than additional transfers from other governments as a source of the revenue gains. I then examine which tax categories contribute to this increase. Sales tax receipts rise by 5.5 percent, and

[5]The event study for data center count does not reject jointly zero pre-treatment coefficients ($p = 0.176$).

[6]One concern is that the subsidies may also benefit Bitcoin mining facilities, which resemble data centers and may use water for cooling. I find no increase in the number of Bitcoin mining facilities following incentive adoption, consistent with the policies distinguishing data centers from cryptocurrency mining operations.

property tax receipts increase by 1.9 percent, while public utility taxes and license taxes do not change significantly. These findings suggest that incentives can expand components of the local tax base even when qualifying data center purchases and property receive exemptions. Finally, I examine revenue from publicly operated utilities and charges for public services. Utility revenue increases by 4.7 percent, whereas I do not find statistically significant changes in parks and recreation, education, or sewerage charges. The increase in utility receipts is consistent with greater demand for publicly provided utilities following incentive adoption.

The revenue gains are accompanied by higher spending on services and amenities for residents. Total expenditure rises by approximately 3.1 percent, but the composition of this spending is particularly informative. Spending on services, including education, parks, public safety, and health, increases by 2.3 percent. Education expenditure rises by 1.8 percent, while education construction spending increases by 8.2 percent, suggesting investment in facilities that serve local communities. I also find a modest improvement in math scores, although no significant change in reading scores. Together, these findings suggest that the fiscal gains are accompanied by investments that benefit residents.

Total long-term debt does not change significantly following incentive adoption. I therefore examine its composition, distinguishing private-purpose debt, which governments issue to finance private-sector activities such as industrial development, private hospitals, and housing, from debt for governmental purposes, which supports public activities such as schools, roads, and utilities. Debt for governmental purposes increases by approximately 5.5 percent, while the estimate for private-purpose debt is negative but statistically insignificant. Together with the expenditure results, this pattern suggests a reorientation of borrowing toward public infrastructure and services rather than a broad expansion of indebtedness. Municipal bond yields and spreads nevertheless rise by approximately 11 and 10 basis points, respectively, suggesting that the fiscal gains coexist with higher borrowing costs, potentially reflecting uncertainty about future revenues and continuing expenditure

commitments.

The labor market results reveal an important employment trade-off associated with data center development. Aggregate effects are modest but negative: exposed counties experience a 1.0% decline in total private employment and a 0.6% decline in average wages. These aggregate declines, however, mask gains in sectors directly connected to data center construction and operation. Nonresidential construction employment increases by 3.9%, while wages in the sector rise by 2.2%. Employment also increases in power generation and natural-gas distribution, with particularly large employment and wage gains in natural-gas distribution. At the same time, residential construction employment declines by 3.2%. These findings suggest that data center development reallocates employment toward directly exposed industries, but that these sectoral gains are insufficient to offset employment losses elsewhere in the local economy.

The employment effects are concentrated in industries that depend heavily on local resources. Subsidy exposure has little detectable effect on employment in non-resource-intensive industries, whereas employment declines significantly in resource-intensive industries. This pattern is consistent with data center development crowding out industries that compete for electricity, water, land, and infrastructure capacity. Combined with employment gains in sectors directly connected to data center construction and operations, the results suggest that subsidies reallocate local economic activity toward data center-related infrastructure and energy sectors rather than generating broad-based employment growth.

Finally, I examine how data center development affects local resource prices. The direction of these effects is ambiguous. Data centers may increase residential electricity prices by raising demand and requiring additional grid investment, but regulated rates and negotiated power contracts may limit the costs passed on to households. Data center development may also increase developed-land prices by creating demand for construction-ready sites. Its effect on agricultural land is less clear: farmland may become more valuable because of its potential conversion to industrial use, or less valuable if data center development raises

concerns about water availability.

I find that data center exposure increases developed-land prices by approximately 19.0%, consistent with greater demand for development-ready property. In contrast, the average effects on residential electricity and agricultural-land prices are small and statistically insignificant.

To answer my second question, whether the local consequences of data center investment vary with project scale, I examine heterogeneity across subsidy adoption cohorts.In 2016, the expansion of cloud computing increased the need for larger data center facilities to store data and provide computing services to businesses (Shehabi et al., 2016). I find evidence consistent with this change in my sample: median facility size increased from approximately 58,000 square feet before 2016 to 250,000 square feet in 2016 or later, while median power capacity rose from 5 MW to 66 MW. I therefore examine heterogeneity across subsidy adoption cohorts, comparing states that adopted subsidies before 2016 with those that adopted them in 2016 or later.

Before 2016, exposure to data center incentives is associated with little change in residential electricity prices or agricultural land values, and no detectable decline in employment. After 2016, however, the same policy exposure is associated with higher residential electricity prices, a sharp decline in agricultural land values, and lower local employment. Developed land values increase in both periods, consistent with stronger demand for sites suitable for data center development. I also examine how data center development affects local elections. Incumbents are more likely to be reelected in states that adopted subsidies before 2016, but less likely in states that adopted them in 2016 or later.

A remaining concern of my empirical design is that tax incentives reflects lobbying by firms that already planned to locate in a state. These location decisions may respond to local economic conditions that also affect the outcomes I examine. To address this concern, I hand collect evidence of lobbying from news reports and local government meeting records and identify states where firms lobbied for data center incentives. I then exclude these

states and find that my main results remain robust.

I also address concerns that data center subsidies are adopted as part of broader incentive packages. Other provisions in these packages may affect local outcomes independently of data center investment. I examine the legislation and timing of each subsidy adoption to identify incentives enacted as part of a larger package. I exclude these states and find that my results remain robust.

I also address concerns that local governments may offer their own subsidies to attract data centers. If these incentives coincide with state subsidy adoption and disproportionately benefit waterfront counties, they could contribute to the estimated effects. Tracking individual local incentives is difficult because disclosure is incomplete and negotiations may involve nondisclosure agreements. To address this concern, I use differences in local government authority under Home Rule and Dillon's Rule. Home Rule generally gives local governments broader discretion over local affairs, whereas Dillon's Rule limits their authority to powers granted by the state or necessarily implied by those powers. I restrict the sample to states whose counties operate under Dillon's Rule, where local policy discretion is more constrained and incentives must rest on state authorization. The positive effects on data center investment, total revenue, taxes, and expenditure remain significant under this restriction, suggesting that the investment and fiscal results persist in states with more limited local government autonomy.

Next, I examine whether the results are driven by major data center hubs or highly populated counties. I exclude Virginia and California to address concerns that the findings reflect the concentration of data centers in these states. I also exclude the 10 and 20 most populated counties to assess whether the results are driven by the largest local economies. My findings are robust to these sample restrictions.

I also examine whether the findings are sensitive to the event window. I restrict the window to four years before and after treatment and then to three years before and after treatment. My results remain robust to both specifications.

Finally, I address concerns that waterfront counties in states adopting data center incentives experience other changes in economic activity at the same time. Expansions in tourism, commercial fishing, or shipping could affect local outcomes and coincide with data center development. To address this concern, I conduct placebo tests using tourism employment, fishing employment, and state waterborne shipping volume. These outcomes capture activity associated with access to water but are not directly targeted by data center tax incentives. I find no clear changes in these outcomes around incentive adoption (Figure A.3).

This paper contributes to the literature on place-based policies and business tax incentives.[7] One strand of this literature studies discretionary, firm-specific subsidies offered by state and local governments to attract individual firms or projects (Greenstone et al., 2010; Greenstone and Moretti, 2003; Kline and Moretti, 2014; Bartik, 2019). This literature commonly models competition for projects as an auction in which jurisdictions bid against one another through increasingly generous subsidy packages. Such competition can generate a race to the bottom and leave the winning jurisdiction with a winner's curse if the subsidy exceeds the local economic benefits generated by the investment (Bartik, 1994; Chava et al., 2024). Whereas this literature primarily examines whether incentives affect firm location, investment, or employment, I build on it by studying the exposure created for municipalities after an incentive attracts a rapidly growing industry. I show that unintended consequences can arise even in the absence of discretionary bidding. The paper therefore introduces a dynamic dimension to the evaluation of place-based incentives: their local consequences depend not only on whether they attract investment, but also on how the targeted industry subsequently grows and changes. The winner's curse may therefore arise not only because a government pays too much to attract a particular project, but also because an initially modest policy exposes governments and residents to large future commitments.

A second strand examines statutory policies adopted at the state or federal level, includ-

[7] Slattery and Zidar (2020) provide an in-depth analysis of different place-based policies.

ing geographically differentiated corporate taxes, payroll taxes, employment credits, and broad investment incentives (Gale et al., 2015; Suárez Serrato and Zidar, 2016; Giroud and Rauh, 2019; Rauh and Shyu, 2024; Ku et al., 2019; Neumark and Grijalva, 2017). Within this second strand, the paper contributes specifically to the literature on investment tax policy. Most of this literature studies accelerated depreciation and expensing provisions and finds that reductions in the cost of capital increase investment (Hall and Jorgenson, 1967; Cummins et al., 1994; House and Shapiro, 2008; Edgerton, 2010; Zwick and Mahon, 2017; Maffini et al., 2019; Ohrn, 2019). Some work also finds that these investment responses increase local employment, consistent with capital and labor being complements (Garrett et al., 2020). I contribute to this literature in three ways. First, I study sales-tax exemptions on capital purchases and property-tax exemptions on installed capital, a relatively understudied form of investment incentive. Second, I show that complementarity between capital and labor at a subsidized project does not guarantee an increase in employment across the local economy. Subsidized investment may raise resource costs, displace incumbent firms, or induce fiscal adjustments, resulting in an overall decline in local employment despite the positive labor demand generated by the project itself. Third, I extend the analysis beyond investment and employment to examine effects on municipal finances and local resource markets. These outcomes are particularly important for sales- and property-tax exemptions because the policies directly affect local revenues while potentially increasing demands on infrastructure and public services.

Finally, this paper contributes to a growing literature on the economic consequences of the digital and AI economy. Existing work has primarily emphasized the effects of artificial intelligence on productivity (Brynjolfsson et al., 2023), labor markets (Babina et al., 2023), and aggregate economic growth (Acemoglu, 2024). Yet the expansion of AI also requires substantial physical infrastructure, including data centers, electricity generation and transmission, cooling systems, and land-intensive server campuses.

Contemporaneous work by Alvarez et al. (2026) and Yue and Zeng (2026) examines

local employment and electricity-market effects of data center development. I complement this work by studying a broader set of local outcomes, including municipal borrowing costs, government revenues and expenditures, land markets, and political outcomes. Together, the results highlight that the expansion of the digital economy is not only a technological transformation but also a local fiscal and resource transformation.

The remainder of the paper is organized as follows. Section 2 provides institutional background on data centers. Section 3 develops the theoretical framework. Section 4 describes the data. Section 5 presents the empirical methodology, and Section 6 reports the results.

2 Institutional Background

Data centers provide a useful setting for four reasons: they are highly capital-intensive, their location is responsive to state tax incentives, geographic differences in county suitability create within-state variation in exposure to these policies, and their scale and resource demands have increased rapidly with technological change. Together, these features allow me to examine both the local effects of investment incentives and how those effects change as the subsidized projects become larger and more resource-intensive.This section lays out relevant institutional details for the empirical setting.

2.1 Prevalence of Data Centers

Data centers are highly capital-intensive and increasingly central to the modern and AI-enabled economy. In the United States, spending on data center construction has risen dramatically over the past decade, making digital infrastructure a key component of aggregate investment. As of January 2026, U.S. spending on data center construction exceeded $2.4 billion per month (Figure A.4)—roughly sixteen times the level observed in early 2014—and installed data center capacity has reached approximately 82 gigawatts (Sam-

borska, 2026). Data centers house the servers, storage systems, and networking equipment required to store, process, and transmit the vast quantities of data generated by businesses and consumers. As cloud computing, artificial intelligence, and digital services expand, demand for data center capacity is expected to grow substantially. Some estimates suggest that global data center capacity demand could increase more than 3.5-fold by 2030 (Figure A.5) (McKinsey & Company, 2025).

2.2 Data Center Subsidies

Data center incentives generally take three forms: full or partial abatements of state sales taxes, local sales taxes, and local property taxes. State sales-tax abatements reduce the taxes paid on qualifying purchases, which may include servers, cooling equipment, electricity, and construction materials, depending on the state. Local sales-tax abatements provide similar relief from taxes imposed by counties and municipalities. Property-tax abatements reduce the recurring taxes paid on the real and personal property associated with a facility, including buildings and installed equipment. Sales-tax abatements therefore subsidize the purchase and replacement of capital, while property-tax abatements reduce the continuing cost of owning that capital.

By 2025, 40 states had adopted some form of data center tax incentive (Table B.2). All 40 states provide either a full or partial abatement of state sales taxes on qualifying data center purchases or impose no state sales tax (Table B.4). Similarly, 39 states provide a full or partial abatement of applicable local sales taxes or impose no local sales tax. Property-tax abatements are less common: eight states, representing 20 percent of adopters, include them in their data center incentive programs. The prevalence of sales-tax abatements is particularly important in this setting because data centers make large and recurring purchases of servers and other equipment. Figure A.6 illustrates how these incentives reduce the tax liability of a qualifying data center.

Most states condition eligibility on the scale of the proposed project. Thirty-six states,

or 90 percent of adopters, impose a minimum investment threshold, while 29 states, or 72.5 percent, impose a minimum employment requirement. When states without a requirement are assigned a value of zero, the mean investment threshold is $110.8 million and the mean employment threshold is 17.3 jobs. Wage requirements are considerably less common: only eight states, or 20 percent, require qualifying projects to meet a wage standard.

These provisions illustrate how data center incentives differ from policies tied primarily to local employment. Eligibility is generally determined by the amount of capital invested, while employment and wage requirements are less common and relatively modest compared with the scale of the qualifying investment. Moreover, the value of the tax abatement can increase with the amount of equipment purchased and property installed without requiring a proportional increase in employment. Data center incentives therefore provide a useful setting for examining how subsidies tied to capital investment affect local employment, public finances, and resource markets.

These incentives can meaningfully affect where data center investment occurs. For example, one study estimates that approximately 90 percent of data center activity in Georgia may be attributable to the state's tax incentives (Justice and Shepherd, 2025). The staggered adoption of these policies provides variation in when states become more attractive to data center investment, providing a useful source of policy-driven variation in exposure to capital-intensive development.

2.3 Geographical Suitability

The setting offers rich within-state geographic variation. Figure 2 maps the water-adjacent and non-water-adjacent counties. A concern with using state tax incentives is that their adoption may coincide with other state-level policy changes or shocks. Even when incentives apply statewide, however, their effect on actual data center investment is concentrated in counties that are suitable for development. In this context, water-adjacent counties are more exposed because large data centers often require access to water resources for cool-

ing. Figure A.7 illustrates this spatial pattern, showing that data centers in the Pacific Northwest and Ohio Valley tend to cluster near major rivers and other water bodies. This creates within-state variation: water-adjacent and non-water-adjacent counties face the same state-level policy environment, but water-adjacent counties are more likely to receive subsidy-induced data center investment. Appendix C provides additional details on other factors that influence data center location.

I leverage the staggered timing of tax incentives and county suitability for data center development to identify the causal impact of attracting capital-intensive.

2.4 Technological Change and Evolution of Data Centers

The scale of data centers has grown dramatically over time. Early-generation facilities were often relatively small operations housed in office buildings or industrial parks (Figure A.8). By contrast, contemporary hyperscale data centers operate on a substantially larger scale, with facilities supporting cloud computing and artificial-intelligence workloads often encompassing hundreds of thousands—or even millions—of square feet, housing extensive server fleets, and requiring electricity capacity measured in hundreds of megawatts. Appendix D provides additional details on the different types of data centers and their operating characteristics.

Major technology firms—including Amazon, Google, Meta, and Microsoft—have sharply increased capital expenditures, much of which is directed toward new data center infrastructure across the United States. Recent announcements illustrate the scale of this shift. Meta's Hyperion data center in Richland Parish, Louisiana, is expected to reach approximately 15 million square feet and consume 5 gigawatts of power at full capacity. This load is roughly half of New York City's peak electricity consumption, underscoring how data center development has shifted from individual facilities toward massive infrastructure campuses (Goldman, 2026).

The facility-level data allow me to document this evolution systematically. Table B.5

reports the cumulative growth of data center development over the sample period, while Figure A.9 presents the corresponding cumulative series graphically. Together, these exhibits document the rapid expansion of the industry's recorded development footprint. However, these cumulative measures show growth in the total number of data centers, not whether the typical facility has become larger.

I therefore examine the evolution of facility scale separately. Table B.5 reports annual median facility square footage and power capacity. Figure A.10 plots these medians over time. Both measures exhibit a pronounced increase after 2016, suggesting that the industry's expansion occurred along both the extensive and intensive margins: more facilities were developed, and the typical facility became larger and more power-intensive. The vertical markers show major developments in the industry, including the acceleration of cloud adoption, the expansion of hyperscale computing, and the emergence of generative-AI demand.

Table B.6 provides a more detailed comparison of facilities before and after 2016. For facility square footage and selected power capacity, the table reports the number of observations, mean, median, standard deviation, and interquartile range in each period. Both the mean and median are higher after 2016, suggesting that the increase is not driven solely by a small number of exceptionally large projects.

I formally assess the post-2016 change by regressing logged facility size and power capacity on a post-2016 indicator, with state fixed effects and state-clustered standard errors. Table B.7 reports the estimates. Figure 3 plots year-specific estimates and 95 percent confidence intervals relative to 2015. These results show that later facilities are larger and more power-intensive, suggesting that later-adopting states may attract projects with different local consequences.

3 Theoretical Motivation

The theoretical framework, developed formally in Appendix E, generates several predictions that guide the empirical analysis. First, subsidies and favorable demand shocks increase investment in data center capacity, but their effects on aggregate employment are ambiguous. Data center expansion creates direct employment and may increase employment in complementary industries. At the same time, greater demand for electricity, water, land, and other locally supplied inputs can raise resource prices and crowd out activity in other sectors.These displacement effects are more likely to arise when data center investment is large relative to local resource capacity, so that the expansion places meaningful pressure on scarce locally supplied inputs. Total employment increases only when the direct and complementary employment gains exceed these displacement effects.

Second, the model predicts substantial heterogeneity across industries and locations. Employment should grow more strongly in industries that complement data center activity, whereas resource-intensive industries should experience weaker employment growth or employment declines. Crowding out should also be more pronounced in locations with inelastic resource supply, limited labor-market slack, or an industrial composition concentrated in resource-intensive sectors. Because data center capital adjusts gradually, these employment and resource-market effects should build over time rather than appear immediately following subsidy adoption.

Third, the fiscal consequences of subsidized investment are theoretically ambiguous. Data center development may expand the local tax base and increase government revenues, but these gains may be accompanied by tax expenditures, infrastructure spending, and additional borrowing. Consequently, higher revenues need not translate into an improved fiscal position and may instead coincide with increases in expenditures and debt.

Finally, the model predicts that the local costs of data center development become more pronounced when realized expansion exceeds policymakers' initial expectations. Large

increases in capacity should generate greater pressure on electricity markets, land prices, and other local resources. Because these costs are unevenly distributed across residents and industries, unexpectedly large expansions may reduce local welfare even when the subsidized project appeared beneficial ex ante. This divergence between expected and realized benefits may also generate political consequences, including greater opposition to incumbent officials.

4 Data

The analysis combines multiple data sources, including employment data from the Quarterly Census of Employment and Wages (QCEW), fiscal data from the Census of Governments Annual Survey of State and Local Government Finances, electricity price data from the U.S. Energy Information Administration (EIA), political data from the American Local Government Elections Database, and a hand-collected dataset of U.S. data center locations. Summary statistics are reported in Table 1, and variable definitions are provided in Table B.1.

4.1 Data Center Data

The data center analysis uses the Aterio U.S. inventory, supplemented with hand-collected information from Baxtel.com to fill in missing facilities and inventory details, including announcement dates, estimated space, and power capacity. I retain records with county identifiers, facility names, stages, and a usable event date. The cleaned inventory contains 5,862 facilities with recorded development dates through 2025, of which 5,855 are matched to the county-year panel.[8]

I aggregate facility records to county-years and construct cumulative counts, facility square footage, and selected power capacity. County-years with no matched inventory

[8]Data center counts may differ across sources because of differences in coverage, reporting dates, and the unit counted, such as individual buildings versus entire campuses. This inventory also includes announced and under-construction projects, whereas some sources count only operating facilities.

event receive zero new activity. The analysis covers 1996–2025, with any earlier activity retained when constructing cumulative stocks. Panel A of Table 1 reports statistics for this period, counting each county-year once. Across all counties and years from 1996 through 2025, the average county-year has 0.325 data centers, 38,760 square feet of facility space, and 9.673 MW of power capacity. These averages include county-years with no recorded data center activity.

4.2 Census of Governments Annual Survey of State and Local Government Finances

To examine the fiscal consequences of data center development, I use data from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. The survey provides annual information on local government revenues, expenditures, and debt, including property and sales taxes, general charges, intergovernmental transfers, current operating expenditures, capital expenditures, construction spending, and debt issuance.

I aggregate these data to the county-year level and merge them with the county-level geographic and treatment data using five-digit county FIPS codes. The resulting panel covers 1995 through 2023 and contains 85,367 matched county-year observations before stacking. Monetary amounts are reported in thousands of nominal dollars.

I construct total sales taxes as the sum of general and selective sales taxes, total expenditures as the sum of direct and intergovernmental expenditures, and capital expenditures as the sum of construction spending and other capital outlays.

Table 1 reports descriptive statistics. The average county-year has total revenue of approximately $376.1 million, with a standard deviation of $977.0 million. Total taxes average $129.5 million, including an average of $98.5 million in property taxes and $16.7 million in sales taxes. Intergovernmental revenue and general charges average $137.2 million and $53.0 million, respectively. On the expenditure side, total expenditures average approximately $165.0 million, while capital expenditures average $63.4 million. Construction

spending and other capital outlays average $36.3 million and $25.7 million, respectively. The large standard deviations relative to the means reflect substantial differences in the size and fiscal activity of counties across the sample.

4.3 Quarterly Census of Employment and Wages (QCEW)

The QCEW is based on administrative records from state unemployment insurance programs and covers approximately 95 percent of all U.S. wage and salary employment. Because it is constructed from mandatory employer filings, the dataset provides near-universal coverage of employment across industries and geographic areas.

The QCEW reports quarterly information on employment, wages, and establishment counts at the county and industry level. Establishments are classified using the North American Industry Classification System (NAICS), enabling measurement of employment dynamics within narrowly defined industries. For this study, data centers are identified using NAICS 518210, which includes establishments primarily engaged in data processing, hosting, and related services.

4.4 U.S. Energy Information Administration (EIA) Data

I incorporate electricity price data from the U.S. Energy Information Administration (EIA). Data centers are highly electricity-intensive facilities, often requiring hundreds of megawatts of power capacity to operate large clusters of servers and cooling systems. Electricity prices therefore play an important role in determining the location and operating costs of data center facilities. The EIA provides detailed data on average electricity prices by customer class and geographic region.

4.5 Acres.com

Land price data are obtained from Acres.com, which provides parcel-level information on land transactions, acreage, land use, and sales prices. The parcel-level data are aggregated

to the county-year level by summing total acres and total transaction values. The average price per acre is then constructed as the total transaction value divided by the total acres.

The data also allow land prices to be measured separately by land type, including developed land, farmland, forest land, water-related land, and open-water categories. These measures are used to examine whether data center subsidies create pressure in local land markets, particularly by increasing the value of developed land or changing the relative value of agricultural and undeveloped land. The county-year land price measures are merged to the main panel using county and state identifiers.

4.6 American Local Government Elections Database

To account for the political environment surrounding local policy decisions, I incorporate election data from the American Local Government Elections Database compiled by De Benedictis-Kessner, Lee, Velez, and Warshaw. This dataset provides detailed information on local elections across the United States, including mayoral, city council, and county-level elections. It includes candidate information, vote totals, election outcomes, and measures of political competition and partisan alignment in local governments.

5 Empirical Strategy

5.1 Identification Strategy

The empirical analysis aims to estimate the local economic and fiscal effects of data center expansion while addressing two related sources of endogeneity. First, states that adopt data center tax incentives may differ systematically from states that do not. Second, within adopting states, data centers are not randomly located across counties. Firms may select locations with better infrastructure, lower land costs, more favorable permitting environments, or stronger expected economic growth. A simple comparison of counties that receive data centers with counties that do not may therefore confound the effects of

data center development with the characteristics that make particular locations attractive to these facilities.

I address these concerns by combining staggered changes in state-level data center tax incentives with predetermined geographic variation in counties' suitability for data center development. Data centers require substantial cooling capacity, and proximity to large water bodies can facilitate cooling and reduce operating costs. Water-adjacent counties are therefore more suitable for large-scale data center development than otherwise comparable inland counties. Consistent with this mechanism, data centers are disproportionately located in counties with access to major water resources. Importantly, proximity to large water bodies is predetermined relative to contemporary state incentive policies and cannot itself respond to subsidy adoption.

Water-adjacent status is constructed using county boundary shapefiles from the U.S. Census Bureau and spatial data on major water bodies. County polygons are matched to water features using geographic information system (GIS) methods. A county is classified as a water-adjacent county if its boundary intersects a major coastline, lake, river, or other large water body. The resulting indicator captures counties with greater geographic suitability for data center development, since large facilities may benefit from access to water resources for cooling, utility infrastructure, and large-scale industrial siting.

The identification strategy uses a difference-in-difference-in-differences design. The first difference compares outcomes before and after the adoption of a state data center incentive. The second compares adopting states with states that have not yet adopted or never adopt an incentive. The third compares water-adjacent counties with inland counties within the same state. The identifying variation therefore comes from changes in outcomes for water-adjacent relative to inland counties following subsidy adoption in treated states, relative to the corresponding within-state difference in untreated states.

Because states adopt data center incentives in different years, I implement the analysis using a stacked difference-in-differences framework following Gormley and Matsa (2011)

and Cengiz et al. (2019). Conventional two-way fixed-effects estimators can produce difficult-to-interpret estimates when treatment timing is staggered and treatment effects are heterogeneous across cohorts or over time (Goodman-Bacon, 2021; de Chaisemartin and d'Haultfoeuille, 2020; Sun and Abraham, 2021; Callaway and Sant'Anna, 2021). The stacked design addresses these concerns by constructing a separate cohort around each treatment event and estimating treatment effects over a common event-time window.

For each year in which a state adopts a data center incentive, I construct a cohort consisting of the states treated in that year and a set of control states that remain untreated over the relevant event window. States that have already adopted a data center incentive are excluded from the control group. Control states therefore consist of never-treated states and, where applicable, states that have not yet adopted an incentive. The cohort-specific samples are then stacked into a single estimation sample.

Because the same county or calendar year may appear in more than one cohort, the fixed effects are interacted with cohort indicators. This allows each cohort to have its own set of fixed effects and prevents repeated observations from receiving inappropriate weight through the fixed-effect estimation.

The baseline specification is:

$$Y_{isct} = \beta\,(Treat_{sc} \times Post_{sct} \times WaterAdjacent_i) + \alpha_{ic} + \lambda_{stc} + \mu_{WaterAdjacent_i \times t \times c} + \varepsilon_{isct}, \quad (1)$$

where Y_{isct} denotes an outcome for county i, located in state s, in year t and cohort c. $Treat_{sc}$ equals one for states belonging to the treated group in cohort c, and $Post_{sct}$ equals one following adoption of the data center incentive. $WaterAdjacent_i$ is an indicator equal to one for counties located adjacent to a major water body.

The coefficient of interest is β. It measures the differential change following subsidy adoption in water-adjacent counties in treated states relative to inland counties in the same state, net of the corresponding water-adjacent–inland difference in untreated states. The

coefficient therefore captures the incremental effect of subsidy adoption in counties that are more geographically suitable for data center development.

County-by-cohort fixed effects, α_{ic}, absorb time-invariant county characteristics within each stacked cohort, including geography, baseline economic conditions, industrial composition, fiscal capacity, and persistent infrastructure differences. State-by-year-by-cohort fixed effects, λ_{stc}, absorb shocks affecting all counties within a state in a given year, including changes in state economic conditions, fiscal policy, energy markets, political conditions, and other state-level policies. I additionally include water-adjacent-by-year-by-cohort fixed effects, $\mu_{WaterAdjacent_i \times t \times c}$, which absorb time-varying shocks common to water-adjacent counties.

Together, these fixed effects ensure that identification comes from differential changes between water-adjacent and inland counties within treated states around subsidy adoption, relative to the corresponding differential in untreated states.

Standard errors are clustered at the state level because treatment is assigned through state-level policy adoption and shocks may be correlated across counties within a state and over time.

5.2 Identification Assumptions

The identifying assumption is that, absent the adoption of data center incentives, the difference in outcomes between water-adjacent and inland counties in treated states would have evolved similarly to the corresponding difference in untreated states. The design therefore does not require treated and untreated states, or water-adjacent and inland counties, to have similar outcome levels. Instead, it requires that subsidy adoption is not accompanied by other shocks that differentially affect water-adjacent counties in treated states.

I provide several pieces of evidence supporting this assumption. First, data centers are disproportionately located in water-adjacent counties, establishing the relevance of geographic suitability. Second, Figure A.1 compares the pre-treatment economic char-

acteristics of adopting and non-adopting states and shows no pronounced sorting across GDP per capita, economic growth, population growth, or employment growth. Finally, the event-study specification below examines whether differential trends emerge prior to subsidy adoption.

5.3 Event Study Specification

To examine treatment dynamics and assess pre-treatment trends, I replace the post-treatment indicator in Equation 1 with indicators for event time:

$$\begin{aligned} Y_{isct} = &\sum_{k \neq 0} \beta_k \left[Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}(t - T_{sc} = k) \right] \\ &+ \alpha_{ic} + \lambda_{stc} + \mu_{WaterAdjacent_i \times t \times c} + \varepsilon_{isct}, \end{aligned} \tag{2}$$

where T_{sc} denotes the treatment year for cohort c, and $\mathbf{1}(t - T_{sc} = k)$ is an indicator equal to one when an observation is k years relative to subsidy adoption. Event time $k = 0$ is omitted and serves as the reference period.

The coefficients β_k trace the differential evolution of outcomes in water-adjacent counties in treated states relative to the corresponding water-adjacent–inland difference in untreated states. Pre-treatment estimates provide a test for differential trends before subsidy adoption, while post-treatment estimates show when the effects emerge and how they evolve over time. The event-study specification includes the same fixed effects and state-level clustering as the baseline model.

For the data center outcomes specifically, I use annual event time centered on the first year after adoption, omit year -1. These timing rules apply to Table 2 and Figure 4.

6 Results

6.1 Data Centers

Prior work shows that tax incentives that reduce the cost of capital can increase business investment (Zwick and Mahon, 2017; Ohrn, 2019). Much of this evidence comes from accelerated depreciation provisions that allow firms to deduct investment expenditures more quickly. Data center incentives operate through a related mechanism: sales-tax exemptions reduce the cost of purchasing equipment, while property-tax exemptions reduce the cost of owning installed capital. These policies may therefore encourage additional development or influence where firms locate their projects. Before examining the consequences for public finances, employment, and resource markets, I begin by examining whether the empirical design captures increases in data center development activity.

This initial analysis is important for interpreting the subsequent results. The empirical design uses incentive adoption and counties' geographic suitability to measure exposure to data center development. If the incentives attract investment to more suitable locations, water-adjacent counties in adopting states should experience greater increases in data center activity relative to the comparison counties.

Table 2 reports the results. Column 1 examines cumulative data center count, while Columns 2 and 3 examine cumulative facility space and power capacity. These outcomes capture different dimensions of development: count measures the number of projects, whereas space and power capacity measure their aggregate physical scale. Each outcome is divided by its corresponding mean across unique county-years from 1996 through 2025. The coefficients therefore express changes relative to a common sample mean rather than percentage changes from each county's own baseline.

Column 1 shows an increase in data center count following incentive adoption ($p < 0.01$). The estimated coefficient of 0.341 corresponds to an increase equal to 34.1 percent of the sample mean. With an average of 0.325 facilities per county-year, this represents

approximately 0.11 additional facilities in more exposed counties relative to the comparison group. Because data centers are discrete and relatively uncommon projects, this estimate describes an average response across counties rather than an increase of the same size in every county.

Columns 2 and 3 show that the increase in project count is accompanied by increases in facility space and power capacity ($p < 0.10$ and $p < 0.05$, respectively). The estimates equal 36.5 percent of average facility space and 70.2 percent of average power capacity, corresponding to approximately 14,136 additional square feet and 6.79 MW of additional capacity. These results suggest that incentives attract more data center projects and expand their overall capacity. The power-capacity response is particularly large relative to its sample mean, making electricity demand an important margin to examine in the subsequent analysis.

Figure 4 presents the annual event study for data center count. Event year 0 is the first calendar year after adoption, and year -1 is the reference period. The post-adoption estimates are positive, consistent with an expansion in recorded development activity. Because the inventory includes announcements and projects under construction, these estimates capture development activity rather than only operating facilities.

Taken together, the results indicate that incentive adoption is followed by an expansion in the number and aggregate capacity of data center projects in more exposed counties.

6.2 Local Public Finances

I examine how data center tax incentives affect local government finances. The estimates compare changes in water-adjacent and nonwater-adjacent counties following state incentive adoption with the corresponding differences in comparison states. The specifications include county, state by year, and water-adjacent by year fixed effects, each interacted with cohort. I estimate the financial outcomes using Poisson pseudo maximum likelihood, which accommodates zero outcomes without requiring a logarithmic transformation.

The fiscal consequences are not obvious ex ante. Data center projects may expand the local property and commercial tax base, but the incentives used to attract them reduce taxes on qualifying investment and activity. Their electricity, water, and infrastructure requirements may also increase public spending. I therefore examine whether revenue gains are accompanied by spending on services and amenities for residents, as well as changes in debt issuance and borrowing costs.

6.2.1 Revenue

Panel A of Table 3 reports the effects on total revenue and its components. Following incentive adoption, water-adjacent counties in adopting states experience increases in total revenue (Column 1; $p < 0.05$) and total taxes (Column 2; $p < 0.05$) relative to the comparison counties. The estimates imply increases of approximately 2.0 and 2.5 percent, respectively. Applying these percentage changes to the county means reported in Table 1 gives approximately $6.3 million in additional annual revenue and $3.2 million in additional annual tax receipts for a county at the respective sample means. General charges and intergovernmental revenue have positive point estimates, but neither is statistically significant (Columns 3 and 4). The evidence thus points more clearly to increased tax collections than to additional service charges or transfers from other governments.

Panel B examines the tax categories underlying this result. General and other selective sales tax receipts increase by approximately 5.5 percent (Column 1; $p < 0.05$), and property tax receipts rise by 1.9 percent (Column 2; $p < 0.10$). I do not find statistically significant changes in public utility taxes or license taxes (Columns 3 and 4). The increases in sales and property tax receipts suggest that incentives can expand components of the local fiscal base despite exempting qualifying data center purchases and property from taxation.

Panel C turns to utility revenue and charges for public services, excluding intergovernmental transfers. Revenue from publicly operated utilities increases by approximately 4.7 percent (Column 1; $p < 0.05$). This measure captures receipts from public water, electric-

ity, gas, and transit systems and is distinct from the public utility taxes examined in Panel B. The increase is consistent with greater demand for publicly provided utilities following incentive adoption. I do not find statistically significant changes in parks and recreation, education, or sewerage charges (Columns 2–4).

Figure 5 examines the timing of these effects. Total revenue estimates are close to zero in the adoption year and the following year, with larger positive estimates appearing subsequently. Tax estimates are also positive throughout the post period, although their magnitude varies across years. Utility revenue and sales tax estimates similarly become more positive in later years. These patterns are consistent with revenue gains developing over time following adoption.[9]

6.2.2 Expenditure

The revenue gains are accompanied by higher public spending. To understand how this spending reaches residents, Table 4 examines aggregate expenditure, capital investment, public services, and education outcomes.

Panel A shows that total expenditure increases by approximately 3.1 percent (Column 1; $p < 0.05$). Direct expenditure, which excludes transfers to other governments, rises by 3.0 percent (Column 2; $p < 0.05$), while current operating expenditure increases by 2.1 percent (Column 3; $p < 0.05$). Intergovernmental expenditure does not change significantly (Column 4). The increase therefore reflects greater direct provision of goods and services rather than higher transfers to other governments.

Panel B shows that the spending response also includes investment in physical assets. Combined capital expenditure increases by approximately 5.6 percent (Column 1; $p < 0.05$), with construction spending rising by 5.5 percent (Column 2; $p < 0.10$) and other capital outlays increasing by 8.8 percent (Column 3; $p < 0.05$). Other capital outlays include purchases such as equipment and land. Interest expenditure does not change

[9] Joint tests of the coefficients for event years -5 through -2 do not reject zero pretreatment coefficients for total revenue ($p = 0.623$), total taxes ($p = 0.281$), utility revenue ($p = 0.425$), or sales taxes ($p = 0.656$).

significantly (Column 4). These results indicate that the spending response extends beyond routine operations to investments that may expand the capacity of public facilities and services.

Panel C provides more direct evidence on the purposes of this spending. Services expenditure increases by approximately 2.3 percent (Column 3; $p < 0.05$). This category includes education, parks, libraries, public safety, health, welfare, and housing. Assistance and subsidies also increase by approximately 21.6 percent (Column 4; $p < 0.01$).[10] Utilities and infrastructure expenditure do not change significantly (Columns 1 and 2). The increase in services expenditure suggests that additional public spending supports services and amenities used by residents.

Panel D examines education in greater detail. Total education expenditure increases by approximately 1.8 percent (Column 1; $p < 0.05$), while education construction expenditure rises by 8.2 percent (Column 2; $p < 0.10$). These findings suggest that the broader increase in public spending includes investment in educational facilities. I also examine student achievement using county averages of Stanford Education Data Archive scores. Math scores increase by approximately 0.012 standard deviations (Column 4; $p < 0.05$), while reading scores do not change significantly (Column 3). These results provide evidence of improved math achievement alongside higher education spending.[11]

Figure 6 presents the corresponding event studies for total expenditure, direct expenditure, current operations, combined capital expenditure, services expenditure, and assistance and subsidies. Total, direct, and operating expenditure estimates generally become more positive in the years following adoption. Capital and services expenditure also show larger positive estimates later in the post period, consistent with spending adjustments developing over time. Assistance and subsidies exhibit a less regular pattern and wider

[10]The assistance and subsidies regression uses 42,024 stacked observations, compared with 271,637 for total expenditure.

[11]Finance specifications use data through 2022, but the usable construction, other capital outlay, and education expenditure samples end in 2021. Overall construction includes conditional zero fills in 2022 that drop out during PPML estimation. The education outcomes use OLS over available years in 2009–2025, excluding unavailable scores in 2020–2021. Reading and math coefficients are measured in standard-deviation units.

confidence intervals.[12]

Together, the expenditure results suggest that the revenue gains are accompanied by spending on public services and facilities that serve residents. These findings connect the fiscal gains to investments in local services and amenities.

6.2.3 Debt and Borrowing Costs

I next examine whether the increases in revenue and expenditure are accompanied by changes in outstanding debt and borrowing costs. Table 5 shows no statistically significant change in total long-term debt outstanding (Column 1). I therefore examine its composition, distinguishing debt issued for private purposes from the remaining category covering other or combined purposes. Private-purpose debt finances private activities through government borrowing arrangements, whereas the other or combined category includes borrowing for government purposes and purposes that are not separately identified.

Debt outstanding in the other or combined category increases by approximately 5.5 percent (Column 2; $p < 0.01$). The private-purpose estimate is negative but statistically insignificant (Column 3), and there is no statistically significant change in short-term debt outstanding (Column 4). These estimates suggest an increase in borrowing within the other or combined category alongside little change in total long-term debt. Together with the expenditure results, this pattern is consistent with greater government activity and spending on public services and facilities.

Municipal borrowing costs also increase. Bond yields rise by 11 basis points (Column 5; $p < 0.05$), while Treasury-adjusted spreads increase by 10.4 basis points (Column 6; $p < 0.10$). The increase in spreads indicates that higher borrowing costs are not explained solely by changes in Treasury rates. It suggests that investors require additional compensation for holding municipal debt in the affected counties, potentially reflecting concerns about

[12] Joint pretreatment tests for event years -5 through -2 yield p-values of 0.633 for total expenditure, 0.624 for direct expenditure, 0.734 for current operations, 0.536 for combined capital expenditure, 0.899 for services expenditure, and 0.801 for assistance and subsidies. These tests do not reject jointly zero pretreatment coefficients.

future spending commitments or reliance on a more concentrated local economic base.

Taken together, the fiscal results suggest that data center incentives expand local revenues and support greater spending on services and facilities used by residents. These changes occur without a statistically significant increase in total long-term debt outstanding, but they are accompanied by higher municipal borrowing costs.

6.3 Employment

Employment growth is one of the primary arguments for offering investment incentives. Data center construction and operation may create jobs in construction, utilities, and related services. At the same time, these facilities require relatively few workers compared with the scale of their investment. Their demand for electricity, land, water, and infrastructure may also raise operating costs for businesses that compete for these resources.

Table 6 reports the employment estimates. The number of establishments declines by approximately 0.4 percent, although the estimate is not statistically significant (Column 1). Total private employment declines by approximately 1.0 percent (Column 2; $p < 0.05$), and average quarterly wages decline by approximately 0.6 percent (Column 3; $p < 0.05$). Figure 7 presents the corresponding event studies. These results suggest that the increase in data center investment does not translate into broader employment or wage gains.

The sectoral estimates show where employment gains occur. Table 7 reports an increase of approximately 4.0 percent in nonresidential construction employment (Column 1; $p < 0.01$) and 2.2 percent in nonresidential construction wages (Column 2; $p < 0.10$). Power employment increases by approximately 8.0 percent (Column 3; $p < 0.10$). Natural gas employment and wages also increase (Columns 5 and 6; $p < 0.10$ and $p < 0.05$, respectively). In contrast, residential construction employment declines by approximately 3.1 percent (Column 7; $p < 0.10$). This pattern suggests that data center development increases demand for workers in industries that support these projects even as aggregate employment falls.

I next examine whether the employment decline is concentrated in industries that rely more heavily on local resources. Table 8 reports an additional employment effect of -0.024 log points for resource-intensive industries relative to other industries. Combining this estimate with the baseline coefficient of -0.002 gives an implied employment decline of approximately 2.6 percent in resource-intensive industries. The direction of these estimates is consistent with data centers competing with other local businesses for resources.

Finally, Table B.8 examines whether these employment changes are accompanied by changes in the number of residents or migration patterns. I find no statistically significant change in population or total net migration per 1,000 residents through 2022 (Panel A), with similar results through 2023 (Panel B).

6.4 Resource Prices

I next examine whether data center development affects local electricity and land prices. Data centers require substantial electricity capacity and large development sites, which may increase demand for these resources. The effect on residential electricity prices depends on how utilities allocate the costs of additional generation and grid investment. Land prices may also respond differently depending on whether land is already developed or remains in agricultural use.

Table 9 reports the full-sample estimates. Residential electricity prices increase by approximately 0.2 percent, but the estimate is not statistically significant (Column 1). Developed land prices increase by approximately 21.9 percent (Column 2; $p < 0.05$), consistent with greater demand for sites suitable for development. Agricultural land prices decline by approximately 0.8 percent, although this estimate is not statistically significant (Column 3). The full-sample results therefore indicate that the clearest price increase occurs in developed land.

Figure 8 presents the electricity-rate event studies, and Figure 9 presents the land-price event studies.

6.5 Housing

Data center development may affect housing costs through two competing channels. Greater demand for land may raise land prices and reduce the land available for residential construction, limiting housing supply and increasing house prices and rents. At the same time, noise and pollution associated with data center facilities may make nearby areas less desirable places to live, reducing housing demand and lowering prices.

Table 10 reports the estimates for median monthly gross rent and median owner-occupied home value using county-level American Community Survey data through 2022. Median gross rent declines by approximately 0.8 percent (Column 1; $p < 0.05$), while median home value declines by approximately 0.9 percent (Column 2; $p < 0.10$).

These findings suggest that downward pressure on housing demand outweighs any upward pressure from competition for land and constraints on residential supply. The declines are consistent with noise and air pollution making residential locations less attractive, alongside the reductions in employment and wages documented above. The contrast between rising developed land prices and falling residential values suggests that data center development can increase the value of land for development while reducing its appeal for residential use.

6.6 Heterogeneity Across Data Center Eras

The scale and operating requirements of data centers have changed considerably over time. Table B.5 and Figure A.9 document the growth in facility size and power capacity, while Table B.7 shows that facilities with inventory events in 2016 or later are substantially larger and have greater power capacity. Larger facilities may place greater demands on local electricity systems and land, changing how data center investment affects residents and businesses.

To examine these differences, Table 11 reports estimates separately for observations

before 2016 and from 2016 onward. Panel A shows no statistically significant change in residential electricity prices or agricultural land prices in the earlier period (Columns 1 and 3). Developed land prices increase by approximately 60.5 percent (Column 2; $p < 0.10$). These estimates suggest that the earlier period is characterized primarily by increased demand for developed sites.

Panel B shows broader changes in resource prices from 2016 onward. Residential electricity prices increase by approximately 0.8 percent (Column 1; $p < 0.10$), and developed land prices increase by approximately 42.6 percent (Column 2; $p < 0.05$). Agricultural land prices decline by approximately 19.2 percent (Column 3; $p < 0.05$). This pattern is consistent with greater demand for developed sites alongside rising electricity costs and declining agricultural land values during the period of larger data center projects.

Employment outcomes also differ across the two periods. The earlier employment estimate is small and statistically insignificant (Panel A, Column 4). From 2016 onward, employment declines by approximately 1.3 percent (Panel B, Column 4; $p < 0.10$). Together with the resource-price results, this suggests that the later period combines greater pressure on local resources with declining aggregate employment.

I also examine local political outcomes. Before 2016, the probability that an incumbent loses office falls by 4.2 percentage points (Panel A, Column 5; $p < 0.05$). From 2016 onward, it rises by 7.6 percentage points (Panel B, Column 5; $p < 0.01$). Incumbents therefore experience improved electoral outcomes in the earlier period and worse outcomes in the later period. This reversal is consistent with residents responding differently as the local costs of data center development become more pronounced.

Taken together, the estimates suggest that the local consequences of data center investment change as facilities become larger and require more resources. The later period is associated with higher residential electricity prices, lower agricultural land values, declining employment, and worse electoral outcomes for incumbents.

6.7 Robustness

Table B.9 in the appendix reports additional checks using shorter event windows, alternative state exclusions, state and year clustering, and exclusions of the largest counties.

6.7.1 Lobbying

A potential concern is that the adoption of data center tax incentives reflects firms' existing plans to locate in a state. Firms may choose a location because of favorable economic conditions and then lobby for subsidies. If these conditions also affect local economic outcomes, the estimated effects could reflect these underlying changes rather than the incentives themselves.

To address this concern, I hand collect evidence of lobbying surrounding subsidy adoption from news reports and local government meeting records. I identify and exclude eleven states with documented evidence of firm lobbying. Table 12, Panel A, reports the results. The increase in data center count is equivalent to 37.2 percent of its sample mean (Column 1; $p < 0.05$). Total revenue increases by approximately 2.4 percent (Column 2; $p < 0.10$), total taxes by 3.4 percent (Column 3; $p < 0.05$), and total expenditure by 3.4 percent (Column 4; $p < 0.10$). Employment declines by approximately 1.5 percent (Column 5; $p < 0.05$). These findings are consistent with the main results and suggest that they are not driven by states with documented firm lobbying.

6.7.2 Broader Incentive Packages

A second concern is that data center subsidies may be adopted as part of broader incentive packages. Other provisions in these packages could affect local investment, employment, and government finances, making it difficult to distinguish their effects from those of data center incentives.

To address this concern, I examine the legislation and timing of each state's subsidy adoption. I exclude thirteen states where data center incentives were enacted as part of

a broader package, retaining states with standalone legislation or no identified broader package. Table 12, Panel B, reports the results. The increase in data center count is equivalent to 22.3 percent of its sample mean (Column 1; $p < 0.10$). Total revenue increases by approximately 2.4 percent (Column 2; $p < 0.10$), and total expenditure increases by 4.7 percent (Column 4; $p < 0.01$). Employment declines by approximately 1.1 percent (Column 5; $p < 0.10$). The total tax estimate remains positive but is not statistically significant (Column 3). The results for investment, revenue, expenditure, and employment therefore persist after excluding states with broader incentive packages.

6.7.3 Local Government Authority

Local governments may also offer their own subsidies to attract data centers. If these incentives coincide with state subsidy adoption and disproportionately benefit water-adjacent counties, they could contribute to the estimated effects. Tracking individual local incentives is difficult because disclosure is incomplete and negotiations may involve nondisclosure agreements.

To address this concern, I use differences in local government authority under Home Rule and Dillon's Rule. Home Rule generally gives local governments broader discretion over local affairs. Dillon's Rule limits their authority to powers granted by the state or necessarily implied by those powers, requiring local incentives to rest on state authorization. Using the National Association of Counties' classifications, I restrict the sample to fourteen states whose counties operate under Dillon's Rule.

Table 12, Panel C, reports the results. The increase in data center count is equivalent to 37.7 percent of its sample mean (Column 1; $p < 0.05$). Total revenue increases by approximately 3.6 percent (Column 2; $p < 0.05$), total taxes by 4.9 percent (Column 3; $p < 0.01$), and total expenditure by 5.3 percent (Column 4; $p < 0.01$). The employment estimate is close to zero and statistically insignificant (Column 5). These findings suggest that the positive investment and fiscal effects persist in states with more limited local

government autonomy.

7 Conclusion

This paper studies the local consequences of state incentives for data center investment and how those consequences change as the underlying technology evolves. I exploit staggered adoption of state-level data center incentives together with predetermined geographic variation in counties' suitability for data center development. Using a stacked difference-in-difference-in-differences design, I compare water-adjacent and inland counties in adopting states around incentive adoption relative to the corresponding difference in untreated states.

The results show that data center incentives have meaningful effects on local resource markets, economic activity, public finances, and political outcomes, but that these effects are not constant over time. In the earlier period, the most visible effect is an increase in developed land values, with comparatively little evidence of pressure on electricity prices, agricultural land, or local employment. After the mid-2010s, however, the pattern changes substantially. Data center exposure is associated with higher residential electricity prices, lower agricultural land values, weaker employment, and greater electoral losses for incumbent politicians. These changes coincide with the industry's transition toward larger, more capital-intensive and resource-intensive facilities.

The findings therefore suggest that technological change can alter the incidence of an investment incentive even when the policy itself does not change. This is particularly important for states that adopted data center incentives before the emergence of hyperscale facilities. These governments designed their policies under an earlier technological regime, yet the incentives continued to apply as the scale, electricity requirements, land demands, and capital intensity of qualifying projects increased. As a result, a policy that initially targeted relatively modest investments could ultimately subsidize projects with substantially different local consequences.

This mechanism has broader implications for the design of place-based and investment incentives. Much of the policy debate focuses on whether governments accurately anticipate the benefits and costs of a proposed investment at the time an incentive is enacted. The evidence in this paper highlights an additional problem: even a policy that is well calibrated to current technology may become poorly calibrated as that technology evolves. Open-ended exemptions, investment-based eligibility rules, and long-lived incentive programs can therefore create fiscal exposure and local externalities that were difficult to anticipate when the policy was adopted.

More broadly, the results caution against evaluating investment incentives solely by the amount of capital they attract. Capital-intensive technologies can generate substantial investment while producing relatively limited employment gains and imposing costs through land, electricity infrastructure, public spending, and other local resources. Whether an investment represents a net benefit to the host jurisdiction therefore depends not only on the amount of capital attracted, but also on how the technology uses local inputs and how those demands evolve over time.

Data centers provide an especially useful setting in which to observe this tension because the industry has undergone rapid technological change while state incentive programs have remained persistent. The findings suggest that policy design should account explicitly for this possibility. Periodic review, expenditure caps, sunset provisions, and eligibility criteria that adjust to changes in project scale or resource intensity may help prevent incentives designed for one technological environment from producing unintended consequences in another. More generally, the paper shows that the effects of economic development policy cannot be understood independently of the technology being subsidized—and that rapidly changing technologies may require equally adaptive policy.

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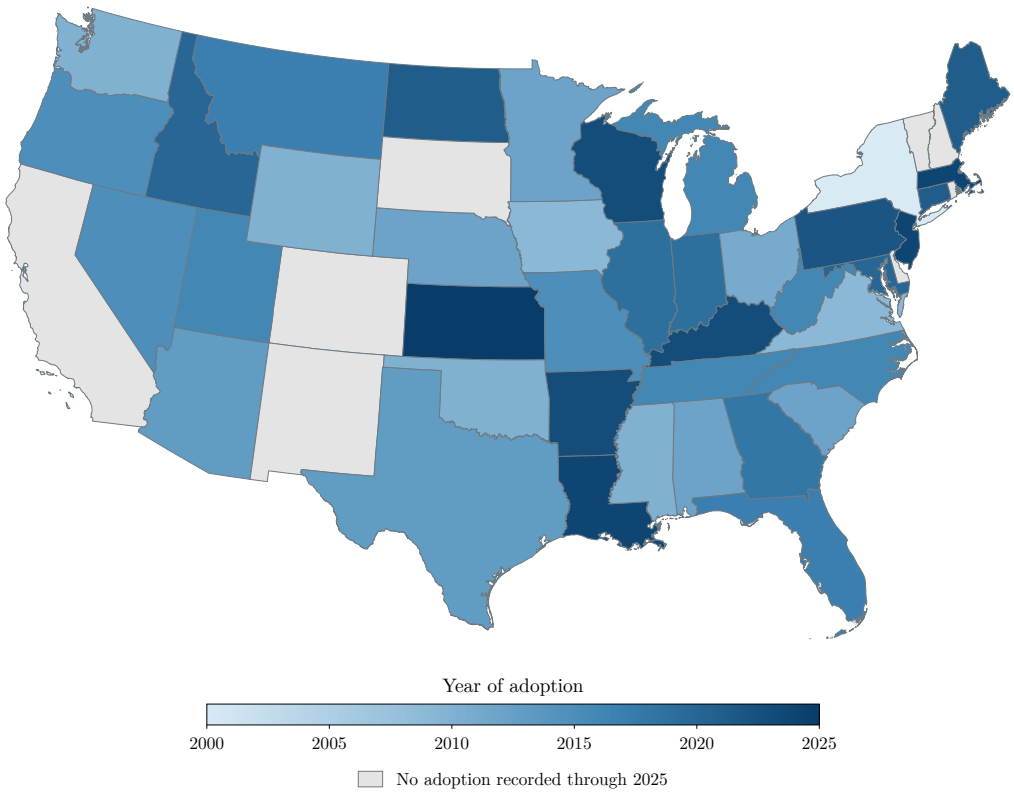
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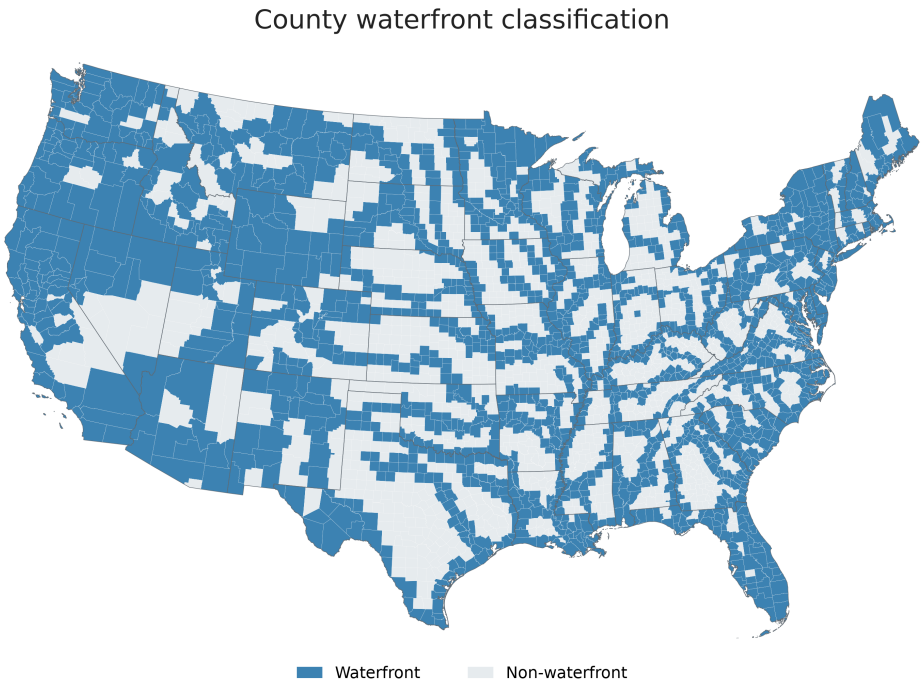
8 Figures

Figure 1. State Adoption of Data Center Tax Incentives



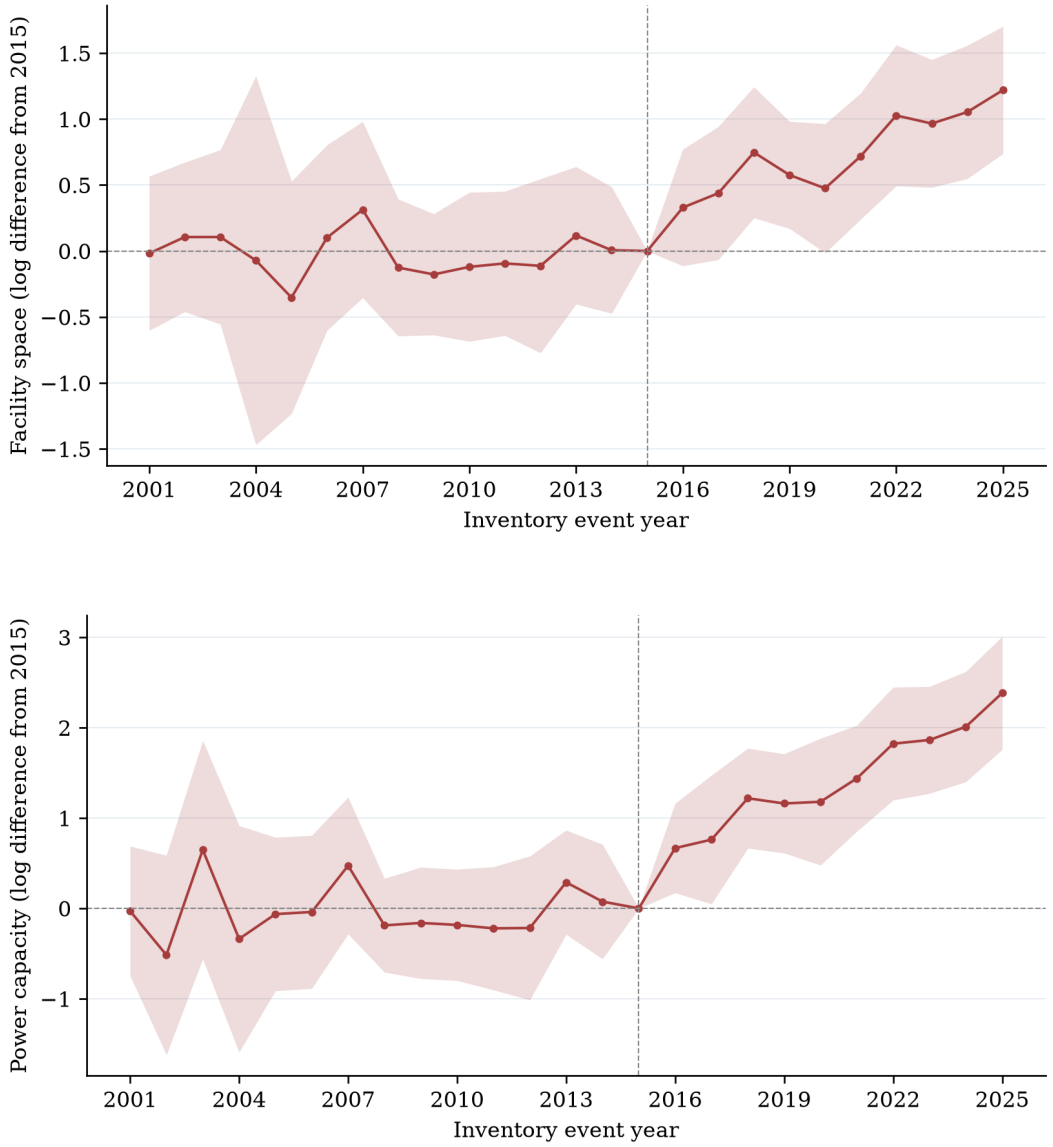
Notes: This figure reports the year in which each state in the United States adopted data center tax incentives, covering adoptions from 2000 through 2025. New York is the earliest adopter in the sample and adopted incentives in 2000. Kansas is the most recent adopter and adopted incentives in 2025. Adoption years are compiled from state legislation and government sources and correspond to those reported in Table B.2. Lighter shades indicate earlier adoption, while darker shades indicate later adoption.Gray identifies states with no recorded adoption through 2025.

Figure 2. Water-Adjacent and Non-Water-Adjacent Counties



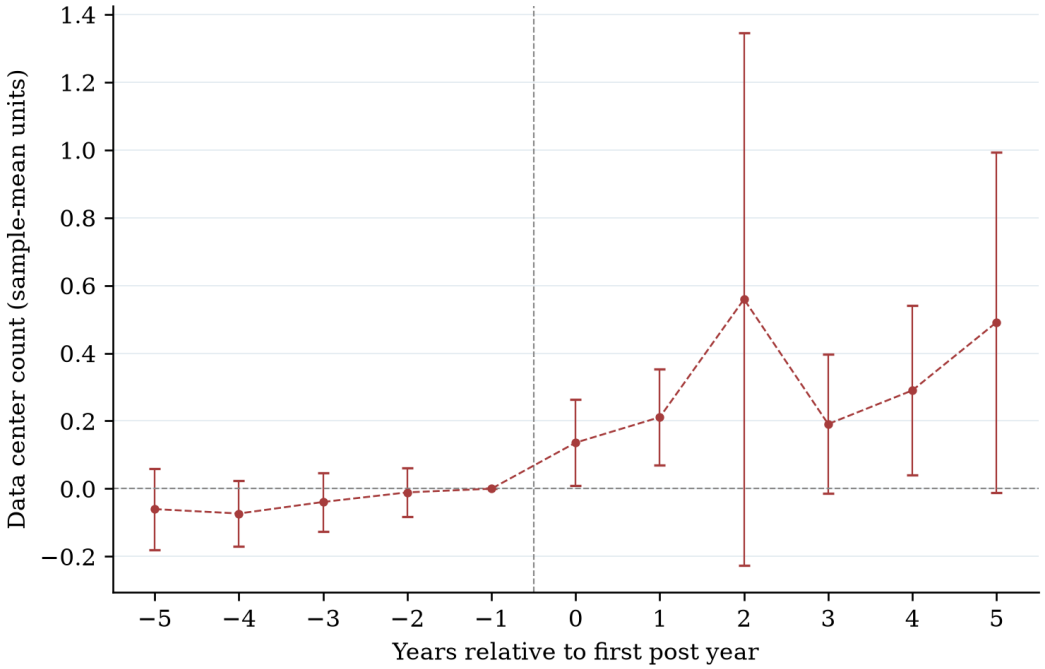
Notes: This figure shows the county water-adjacent classification used in the analysis. Blue identifies water-adjacent counties and light gray identifies non-water-adjacent counties. County boundaries are from the U.S. Census Bureau's 2024 cartographic boundary files.

Figure 3. Evolution of Data Center Scale Relative to 2015



Notes: This figure shows how data center size and power capacity evolve relative to data centers in 2015. The upper panel reports coefficients from a regression of log facility size measured in square footage. The lower panel reports coefficients for log power capacity measured in megawatts. Both regressions include state fixed effects. The sample includes facilities from 2000 through 2025. Event dates use the announcement date when available, followed by construction start, construction completion, and activation when preceding dates are missing. The omitted reference year is 2015. The shaded area represents the 95 percent confidence intervals. Facility data are from the Aterio U.S. inventory and supplemented with information hand collected from Baxtel.com. Standard errors are clustered at the state level.

Figure 4. Dynamic Effects on Normalized Data Center Count

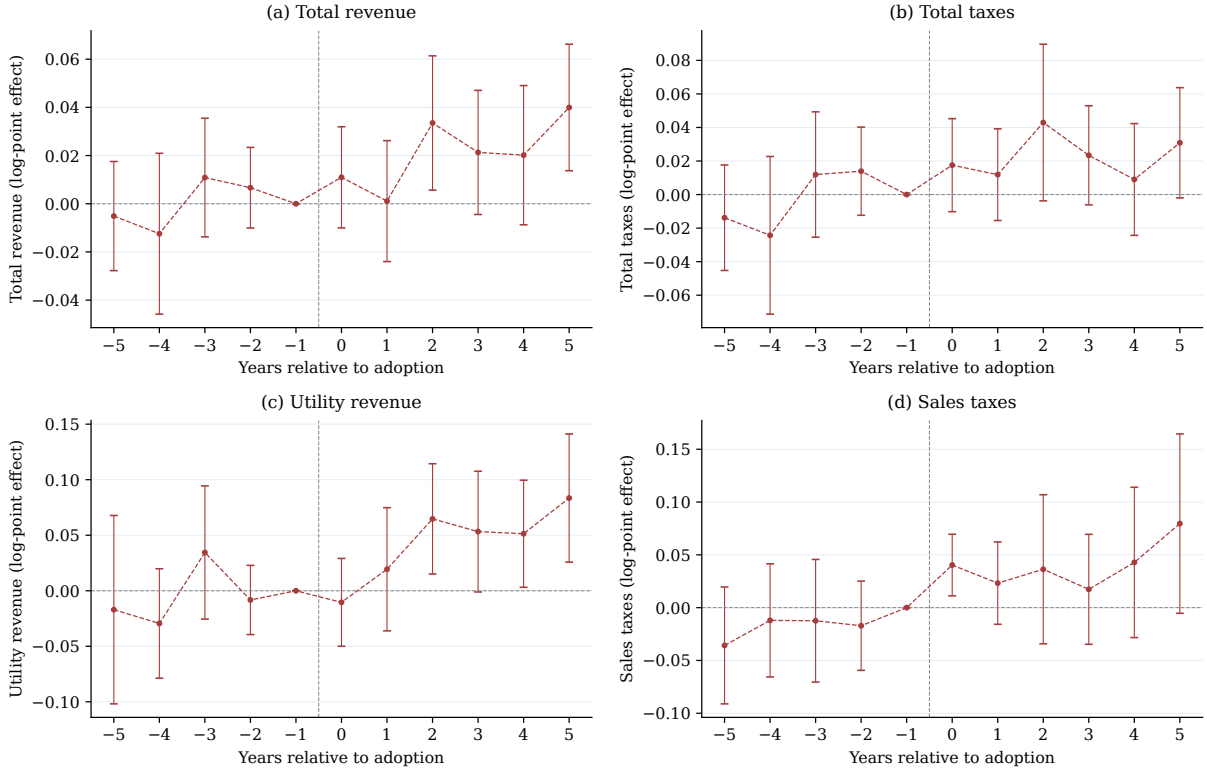


Notes: This figure presents dynamic effects on cumulative data center counts, normalized by the mean across unique county-years from 1996 through 2025. The figure plots the estimated coefficients (β_k) and the 95% confidence intervals from the following stacked triple-difference regression:

$$Y_{isct} = \sum_{\substack{k=-5 \\ k\neq -1}}^{5} \beta_k \cdot (Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{t - T_{sc} = k\}) + \alpha_{ic} + \lambda_{stc} + \mu_{WaterAdjacent_i \times t \times c} + \varepsilon_{isct}.$$

Y_{isct} represents the cumulative number of data centers in county i and year t in cohort stack c normalized by the 1996–2025 mean. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{t - T_{sc} = k\}$ is an indicator for each event year k, where T_{sc} is the calendar year of incentive adoption. Event year -1 is excluded as the reference year. The regression includes county-by-cohort fixed effects (α_{ic}), state-by-year-by-cohort fixed effects (λ_{stc}), and water-by-year-by-cohort fixed effects ($\mu_{WaterAdjacent_i \times t \times c}$). The event window spans five years before through five years after the first post year. The data uses county-year observations from 1996-2025. The joint test that the coefficients for event years -5 through -2 equal zero is not rejected ($p = 0.176$). The corresponding pooled estimate is reported in Column 1 of Table 2. Facility data are from the Aterio U.S. inventory and supplemented with information hand collected from Baxtel.com. Standard errors are clustered at the state level.

Figure 5. Dynamic Effects of Data Center Tax Incentives on Government Revenue

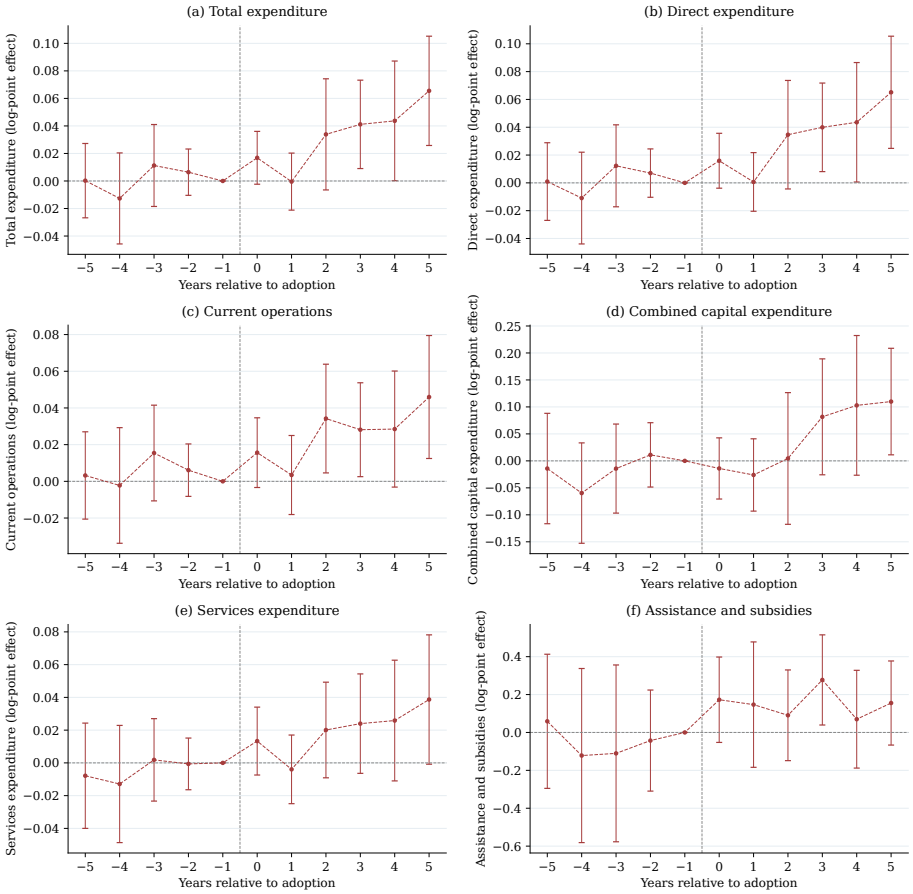


Notes: This figure presents dynamic effects on total revenue (Panel a), total taxes (Panel b), utility revenue (Panel c), and sales tax (Panel d). The figure plots the estimated coefficients (β_k) and the 95% confidence intervals from the following stacked triple-difference regression estimated using PPML:

$$\mathbb{E}[Y_{isct} \mid X] = \exp\left[\sum_{\substack{k=-5 \\ k\neq -1}}^{5} \beta_k \cdot (Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{t - T_{sc} = k\}) + \alpha_{ic} + \lambda_{stc} + \mu_{WaterAdjacent_i \times t \times c}\right].$$

Y_{isct} represents the revenue outcome in county i and year t in cohort stack c. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{t - T_{sc} = k\}$ is an indicator for each event year k, where T_{sc} is the calendar year of incentive adoption. Event year -1 is excluded as the reference year. The regression includes county-by-cohort fixed effects (α_{ic}), state-by-year-by-cohort fixed effects (λ_{stc}), and water-by-year-by-cohort fixed effects ($\mu_{WaterAdjacent_i \times t \times c}$). The event window spans five years before through five years after the first post year. The data uses county-year observations from 1996-2022. The joint test that the coefficients for event years -5 through -2 equal zero is not rejected in any panel ($p = 0.623$, 0.281, 0.425, and 0.656 for Panels (a)–(d), respectively). The corresponding pooled estimates are reported in Table 3. Government revenue data is collected from the Census Annual Survey of State and Local of State and Local Governments.Standard errors are clustered at the state level.Standard errors are clustered at the state level.

Figure 6. Dynamic Effects of Data Center Tax Incentives on Government Expenditure

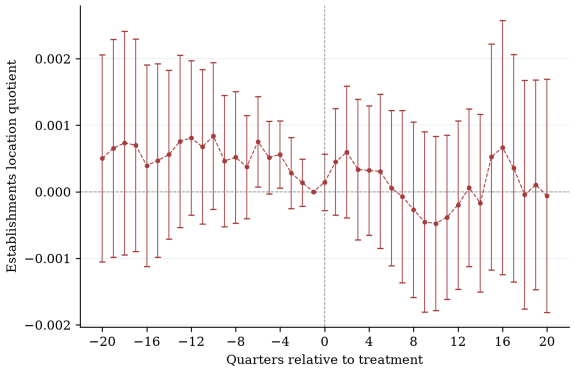


Notes: This figure presents dynamic effects on total expenditure (Panel a), direct expenditure (Panel b), current operations (Panel c), combined capital expenditure (Panel d), services expenditure (Panel e), and assistance and subsidies (Panel f). The figure plots the estimated coefficients (β_k) and 95% confidence intervals from the following stacked triple-difference regression estimated using PPML:

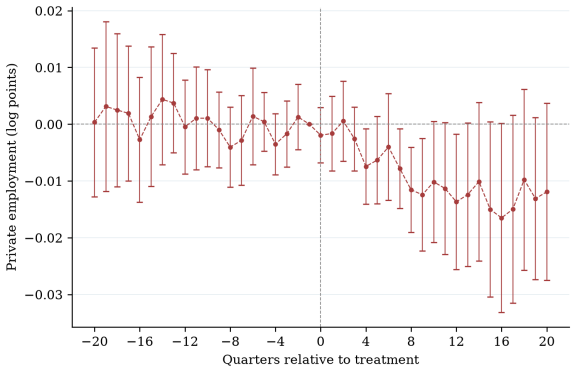
$$\mathbb{E}[Y_{isct} \mid X] = \exp\left[\sum_{\substack{k=-5 \\ k\neq -1}}^{5} \beta_k \left(Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{t - T_{sc} = k\}\right) + \alpha_{ic} + \lambda_{stc} + \mu_{WaterAdjacent_i \times t \times c}\right].$$

Y_{isct} represents the expenditure outcome in county i and year t in cohort stack c. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{t - T_{sc} = k\}$ is an indicator for each event year k, where T_{sc} is the calendar year of incentive adoption. Event year -1 is excluded as the reference year. The regression includes county-by-cohort fixed effects (α_{ic}), state-by-year-by-cohort fixed effects (λ_{stc}), and water-by-year-by-cohort fixed effects ($\mu_{WaterAdjacent_i \times t \times c}$). The event window spans five years before through five years after the first post year. The data uses county-year observations from 1996-2022. The joint test that the coefficients for event years -5 through -2 equal zero is not rejected in any panel ($p = 0.633$, 0.624, 0.734, 0.536, 0.899, and 0.801 for Panels (a)–(f), respectively). The corresponding pooled estimates are reported in Table 4. Government expenditure data is collected from the Census Annual Survey of State and Local of State and Local Governments. Standard errors are clustered at the state level.

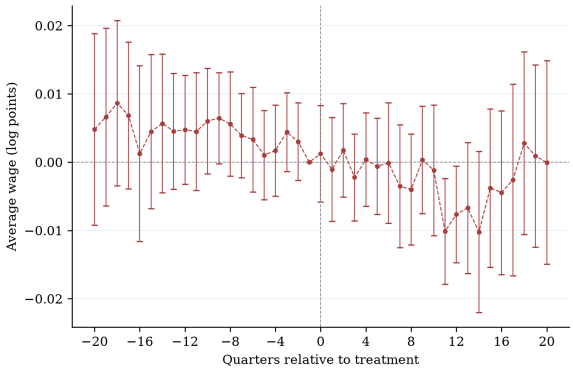
Figure 7. Event Study Estimates for Establishments, Employment, and Wages



(a) Log Total Establishments



(b) Log Total Employment



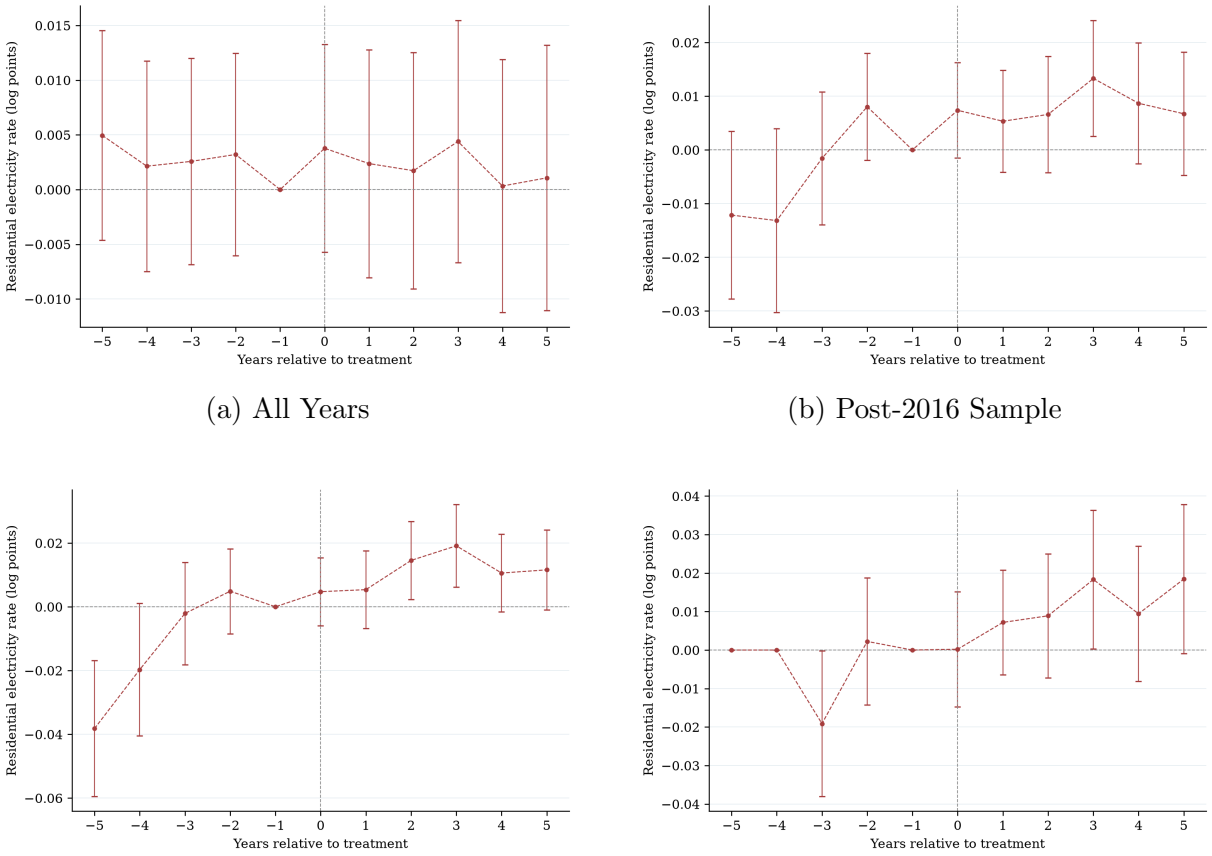
(c) Log Average Wage

Notes: This figure presents dynamic effects on the natural log of number of establishments (Panel a), total employment (Panel b), and average wages (Panel c). The figure plots the estimated coefficients (β_k) and 95% confidence intervals from the following stacked triple-difference regression:

$$Y_{iscq} = \sum_{\substack{k \in \mathcal{K} \\ k \neq -1}} \beta_k \cdot (Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{q - T_{sc} = k\}) + \alpha_{ic} + \tau_{qc} + \lambda_{sqc} + \mu_{WaterAdjacent_i \times q \times c} + \varepsilon_{iscq}.$$

Y_{iscq} represents outcomes in county i and quarter q in cohort stack c. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{q - T_{sc} = k\}$ is an indicator for each event quarter k, where T_{sc} is the treatment quarter and $\mathcal{K}$ contains the quarters in the event window. Event quarter -1 is excluded as the reference quarter. All regressions include county-by-cohort fixed effects (α_{ic}), quarter-by-cohort fixed effects (τ_{qc}), and state-by-quarter-by-cohort fixed effects (λ_{sqc}) and water-by-quarter-by-cohort fixed effects ($\mu_{WaterAdjacent_i \times q \times c}$). The event window spans five years before through five years after the first post year. The data uses county-year observations from 1996-2022. The corresponding pooled estimates are reported in Table 6. Data is from the Quarterly Census of Employment and Wages. Standard errors are clustered at the state level.

Figure 8. Event Study Estimates for Residential Electricity Rates



(a) All Years

(b) Post-2016 Sample

(c) Post-2018 Sample

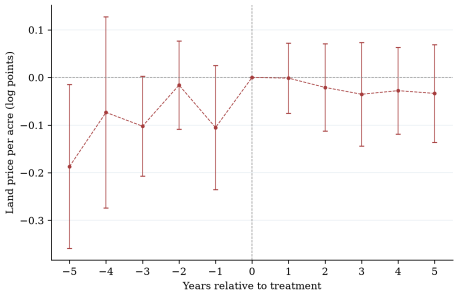
(d) Post-2020 Sample

Notes: This figure presents dynamic effects on log residential electricity rates for all years (Panel a), the post-2016 sample (Panel b), the post-2018 sample (Panel c), and the post-2020 sample (Panel d). The figure plots the estimated coefficients (β_k) and 95% confidence intervals from the following stacked triple-difference regression:

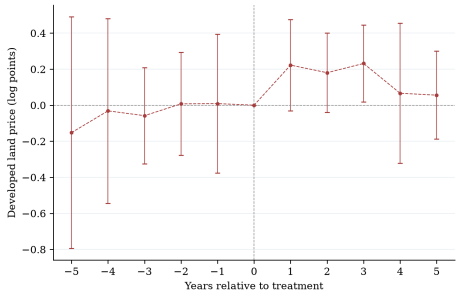
$$Y_{isct} = \sum_{\substack{k \in \mathcal{K} \\ k \neq -1}} \beta_k \cdot (Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{t - T_{sc} = k\}) + \alpha_{ic} + \tau_{tc} + \lambda_{stc} + \mu_{WaterAdjacent_i \times t \times c} + \varepsilon_{isct}.$$

Y_{isct} represents log residential electricity rates in county i and year t in cohort stack c. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{t - T_{sc} = k\}$ is an indicator for each event year k, where T_{sc} is the calendar year of incentive adoption and $\mathcal{K}$ contains the years in the event window. Event year -1 is excluded as the reference year. The regression includes county-by-cohort fixed effects (α_{ic}), year-by-cohort fixed effects (τ_{tc}), state-by-year-by-cohort fixed effects (λ_{stc}), and water-by-year-by-cohort fixed effects ($\mu_{WaterAdjacent_i \times t \times c}$). The event window spans five years before through five years after the first post year. The data uses county-year observations from 2001-2024. The corresponding pooled estimates are reported in Table 9 and Table 11. Data is from the Quarterly Census of Employment and Wages. Standard errors are clustered at the state level.

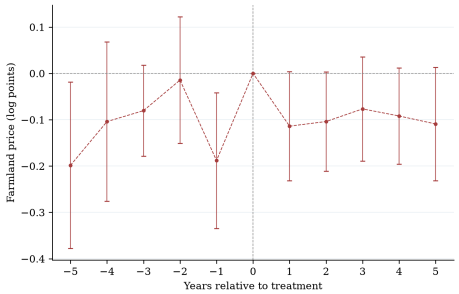
Figure 9. Dynamic Effects of Data Center Exposure on Land Prices



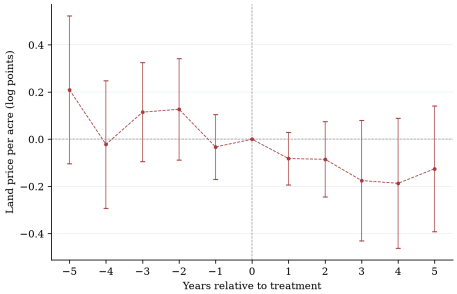
(a) Price per Acre: All Years



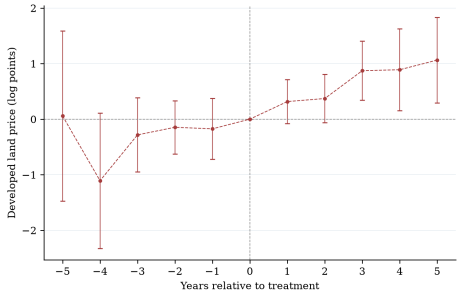
(b) Developed Land Price: All Years



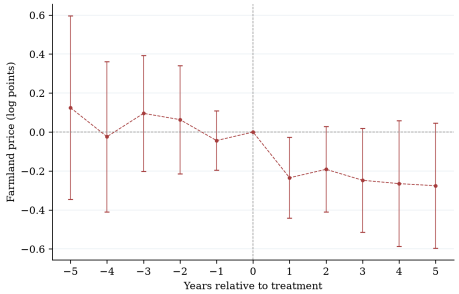
(c) Farmland Price: All Years



(d) Price per Acre: After 2016



(e) Developed Land Price: After 2016



(f) Farmland Price: After 2016

Notes: This figure presents dynamic effects on log land prices per acre (Panels a and d), log developed land prices (Panels b and e), and log farmland prices (Panels c and f). Panels a, b, and c use the full sample, while Panels d, e, and f restrict the sample to years after 2016. The figure plots the estimated coefficients (β_k) and 95% confidence intervals from the following stacked triple-difference regression:

$$Y_{isct} = \sum_{\substack{k=-5 \\ k\neq 0}}^{5} \beta_k \cdot (Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{t - T_{sc} = k\}) + \alpha_{ic} + \tau_{tc} + \lambda_{stc} + \mu_{WaterAdjacent_i \times t \times c} + \varepsilon_{isct}.$$

Y_{isct} represents the log land price outcome in county i and year t in cohort stack c. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{t - T_{sc} = k\}$ is an indicator for each event year k, where T_{sc} denotes the treatment year. Event year 0 is excluded as the reference year. The regression includes county-by-cohort fixed effects (α_{ic}), year-by-cohort fixed effects (τ_{tc}), state-by-year-by-cohort fixed effects (λ_{stc}), and water-by-year-by-cohort fixed effects ($\mu_{WaterAdjacent_i \times t \times c}$). The event window spans five years before through five years after the first post year. The data uses county-year observations from 2010-2024. The corresponding pooled estimates are reported in Table 9 and Table 11. Data is from the Quarterly Census of Employment and Wages. Standard errors are clustered at the

9 Tables

Table 1
Summary Statistics

Variable	Observations	Mean	Median	SD
Panel A: Data Center Investment				
Cumulative number of data centers (count)	94,320	0.325	0.000	3.381
Cumulative facility space (square feet)	94,320	38,760.14	0.00	570,569.58
Cumulative power capacity (MW)	94,320	9.673	0.000	180.390
Panel B: Government Revenue ($ thousands)				
Total revenue	71,400	310,061.69	61,618.50	1,230,529.80
Total taxes	71,427	129,559.31	18,488.00	571,459.16
General charges (including liquor)	71,402	52,840.53	6,974.50	231,140.81
Intergovernmental revenue	71,420	127,575.94	31,811.50	477,567.26
General + other selective sales taxes	71,427	16,491.78	547.00	99,094.83
Property taxes	71,423	97,268.80	14,724.00	385,380.39
Public utility taxes	60,886	3,150.38	85.00	20,754.42
License taxes	66,229	2,934.48	84.00	15,693.80
Utility revenue	64,588	31,372.94	2,449.00	148,539.13
Parks and recreation charges	62,486	1,966.04	57.00	12,138.82
Education charges	71,158	6,150.48	1,107.00	19,039.18
Sewerage charges	63,862	10,578.69	844.50	41,457.69
Panel C: Government Expenditure ($ thousands)				
Total expenditure	71,427	358,085.53	68,599.00	1,430,439.77
Direct expenditure	71,427	349,576.72	67,075.00	1,383,313.39
Current operations	71,427	284,673.14	57,703.00	1,079,500.92
Intergovernmental expenditure	69,431	8,753.42	744.00	63,962.49
Combined capital	71,383	47,797.11	6,489.00	221,307.94
Construction	70,805	35,628.85	3,676.00	175,419.93
Other capital outlays	69,936	11,376.17	2,073.00	50,484.96
Interest on debt	70,801	16,358.55	1,438.00	91,251.06
Utilities expenditure	64,788	40,204.85	2,726.00	220,434.36
Infrastructure expenditure	67,128	34,958.01	5,096.00	153,904.48
Services expenditure	71,416	225,969.71	49,577.50	821,689.47
Assistance and subsidies	47,752	1,377.22	0.00	25,726.79
Total education expenditure	69,839	149,942.85	39,020.00	463,628.24
Education construction	68,218	12,161.13	1,435.00	47,345.89
Panel D: Government Debt ($ thousands)				
Total long-term debt outstanding	70,740	364,341.89	33,580.00	1,905,128.71
Long-term debt outstanding, other/combined	70,732	310,541.68	29,886.00	1,706,452.32
Private-purpose long-term debt outstanding	52,979	71,883.41	0.00	389,865.74
Short-term debt outstanding	54,980	7,970.39	0.00	53,725.06

Table 1 (continued)

Variable	Observations	Mean	Median	SD
Panel E: Employment				
Quarterly establishments (count)	337,792	2,586.76	564.00	10,514.36
Average employment (workers)	337,792	34,812.66	6,274.17	128,854.60
Average wage ($ per quarter)	336,549	8,228.42	7,768.24	2,940.43
Panel F: Electricity Rates ($/kWh)				
Residential electricity rate	72,779	0.1113	0.1089	0.0349
Panel G: Land Prices ($ per acre)				
Overall land	27,512	37,607.65	7,667.88	406,026.43
Developed land	20,067	187,993.73	56,185.98	1,167,651.65
Farmland	27,012	20,052.16	7,862.03	393,427.24
Panel H: Municipal Bond Yield (percentage points)				
Volume-weighted yield	10,806,527	2.5402	2.4575	1.4242
Panel I: Education Outcomes (SD units)				
Reading scores	30,334	-0.055	-0.031	0.258
Math scores	29,904	-0.057	-0.038	0.293

Notes: This table reports the number of observations, mean, median, and standard deviation for the variables before stacking. Panel A uses county-year observations from 1996–2025 on data center investment from the Aterio U.S. inventory and is supplemented with information hand collected from Baxtel.com. Panels B–D use county-year observations from 1996–2022 on government revenue, expenditure, and year-end debt outstanding from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. Other capital outlays and the two education expenditure measures are available through 2021. Revenue, expenditure, and debt are reported in thousands of nominal dollars. Panel E uses county-quarter observations from 1996–2022 on establishments, employment, and wages from the Bureau of Labor Statistics' Quarterly Census of Emloyment and Wages (QCEW). Panel F uses county-year observations from 2001–2024 on residential electricity rates from the U.S. Energy Information Administration (EIA), measured in dollars per kilowatt-hour. Panel G uses county-year observations from 2010–2025 on land prices from Acres.com, measured in dollars per acre. Panel H uses bond-month observations from 2005–2023 on municipal bond yields from the Municipal Securities Rulemaking Board (MSRB), with volume-weighted yields measured in percentage points. Panel I uses county-year observations from 2009–2025 on reading and math scores from the Stanford Education Data Archive (SEDA), excluding unavailable scores in 2020–2021. Scores are measured in standard-deviation units.

Table 2
Stacked Difference-in-Differences Estimates for Normalized Data Center Outcomes

	Normalized count	Normalized space	Normalized power
Post × Water-adjacent × Treated	0.341***	0.365*	0.702**
	(0.119)	(0.193)	(0.341)
County FE	Yes	Yes	Yes
State × year FE	Yes	Yes	Yes
Water-adjacent × year FE	Yes	Yes	Yes
Observations	293,180	291,458	293,180
R^2	0.846	0.759	0.304

Notes: This table reports coefficients from annual stacked county-panel regressions of data center investment on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for years after incentive adoption (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-adjacent-by-year fixed effects. All fixed effects are interacted with cohort indicators. The dependent variables are cumulative data center count (Column 1), facility space in square feet (Column 2), and power capacity in megawatts (Column 3), each divided by its mean across unique county-years from 1996–2025. Each cohort's sample spans five years before through five years after the first calendar year following adoption, with Post equal to one beginning in the year of treatment. Comparison states are never treated or not yet treated at cohort adoption. The data uses county-year observations from 1996-2025. Facility data are from the Aterio U.S. inventory and supplemented with information hand collected from Baxtel.com. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$,and $^{***}p < 0.01$.

Table 3
Effects of Data Center Tax Incentives on Government Revenue

	(1)	(2)	(3)	(4)
Panel A: Revenue and Its Components	Total revenue	Total taxes	General charges	Intergov. revenue
Post × Water-adjacent × Treated	0.0203**	0.0247**	0.0434	0.0094
	(0.0100)	(0.0119)	(0.0304)	(0.0095)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.992	0.991	0.975	0.991
Observations	271,501	271,637	271,512	271,596
Panel B: Tax Revenue	Sales taxes	Property taxes	Public utility taxes	License taxes
Post × Water-adjacent × Treated	0.0540**	0.0190*	-0.0145	0.0079
	(0.0268)	(0.0115)	(0.0447)	(0.0367)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.979	0.992	0.975	0.967
Observations	248,917	271,621	193,314	235,525
Panel C: Utility Revenue and Service Charges	Utility revenue	Parks and recreation charges	Education charges	Sewerage charges
Post × Water-adjacent × Treated	0.0458**	0.0564	0.0058	0.0157
	(0.0219)	(0.0393)	(0.0139)	(0.0265)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.981	0.968	0.987	0.976
Observations	231,185	208,895	269,882	226,695

Notes: This table reports coefficients from annual stacked county-panel PPML regressions of government revenue on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the adoption year and subsequent years (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-adjacent-by-year fixed effects. All fixed effects are interacted with cohort indicators. Total revenue equals taxes, general charges (including liquor), and intergovernmental revenue. Sales taxes comprise general and other selective sales taxes. Each cohort's sample spans five years before through five years after adoption, with Post equal to one beginning in the year of treatment. Comparison states are never treated or not yet treated at cohort adoption. The data uses county-year observations from 1996-2022. Revenue data are from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 4
Effects of Data Center Tax Incentives on Expenditure

	(1)	(2)	(3)	(4)
Panel A: Overall expenditure	Total expenditure	Direct expenditure	Current operations	Intergov. expenditure
Post × Treated × Water-adjacent	0.0308**	0.0298**	0.0206**	-0.0350
	(0.0123)	(0.0123)	(0.0099)	(0.0360)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Observations	271,637	271,637	271,637	256,868
Pseudo R^2	0.992	0.992	0.993	0.961
Panel B: Capital and financing	Combined capital	Construction	Other capital outlays	Interest on debt
Post × Treated × Water-adjacent	0.0549**	0.0539*	0.0847**	-0.0100
	(0.0262)	(0.0314)	(0.0368)	(0.0348)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Observations	271,406	260,723	263,682	267,635
Pseudo R^2	0.964	0.951	0.943	0.983
Panel C: Services and assistance	Utilities expenditure	Infrastructure expenditure	Services expenditure	Assistance and subsidies
Post × Treated × Water-adjacent	0.0362	0.0128	0.0226**	0.1955***
	(0.0322)	(0.0268)	(0.0106)	(0.0742)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Observations	232,664	247,834	271,593	42,024
Pseudo R^2	0.980	0.973	0.992	0.937
Panel D: Education expenditure and learning outcomes	Total education expenditure	Education construction	Reading scores	Math scores
Post × Treated × Water-adjacent	0.0176**	0.0786*	0.0040	0.0118**
	(0.0089)	(0.0448)	(0.0044)	(0.0057)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Observations	263,050	252,633	172,496	173,022
Pseudo R^2	0.992	0.880	–	–
R^2	–	–	0.910	0.904

Notes: This table reports coefficients from annual stacked county-panel PPML regressions of government expenditure and OLS regressions of education outcomes on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the adoption year and subsequent years (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-adjacent-by-year fixed effects, all interacted with cohort indicators. Total expenditure includes intergovernmental transfers, while direct expenditure excludes them. Each cohort's sample spans five years before through five years after adoption, with Post equal to one beginning in the year of treatment. Reading and math scores cover 2009–2025 and are from the Stanford Education Data Archive (SEDA). Comparison states are never treated or not yet treated at cohort adoption. Expenditure data covers 1996-2022 and is from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 5
Effects of Data Center Tax Incentives on Government Debt and Municipal Borrowing Costs

	Debt outstanding				Municipal borrowing costs	
	Long-term: total	Long-term: other/combined	Long-term: private purpose	Short-term	Bond yield	Treasury-adjusted yield spread
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Treated × Water-adjacent	0.0097	0.0535***	-0.1114	-0.0798	0.110**	0.104*
	(0.0234)	(0.0187)	(0.0726)	(0.1423)	(0.050)	(0.053)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	0.986	0.987	0.952	0.937	—	—
Observations	267,302	267,226	111,152	141,825	46,878,051	46,878,051

Notes: This table reports coefficients from annual stacked county-panel PPML regressions of government debt and monthly OLS regressions of municipal bond yields and yield spreads on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the post-adoption period (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-time, and water-adjacent-by-time fixed effects, all interacted with cohort indicators. Time denotes years for the debt regressions and months for the municipal bond regressions. Debt measures year-end outstanding balances. Municipal bond yields and Treasury-adjusted yield spreads are measured in percentage points. Each debt cohort's sample spans five years before through five years after adoption, with Post equal to one beginning in the year of treatment. Comparison states are never treated or not yet treated at cohort adoption. The data uses county-year observations from 1996-2022. Revenue data are from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. Municipal bond data are from the Municipal Securities Rulemaking Board (MSRB). Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 6
Effect of Data Center Exposure on Local Employment

	(1) Establishments	(2) Employment	(3) Average wage
Post × Treated × Water-adjacent	−0.004	−0.010**	−0.006**
	(0.003)	(0.005)	(0.003)
County × Cohort FE	Yes	Yes	Yes
State × Quarter × Cohort FE	Yes	Yes	Yes
Water-adjacent × Quarter × Cohort FE	Yes	Yes	Yes
Observations	1,489,397	1,483,920	1,483,920
R^2	0.999	0.998	0.947

Notes: This table reports coefficients from annual stacked county-panel OLS regressions of establishments, employment, and wages on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the adoption year and subsequent years (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-adjacent-by-year fixed effects. All fixed effects are interacted with cohort indicators. Each cohort's sample spans five years before through five years after adoption, with Post equal to one beginning in the year of treatment. Comparison states are never treated or not yet treated at cohort adoption. The data uses county-year observations from 1996-2022. The data is from the Quarterly Census of Employment and Wages. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 7
Effect of Data Center Exposure on Sectoral Employment and Wages

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Nonres. Const. Emp.	Nonres. Const. Wage	Power Emp.	Power Wage	Natural Gas Emp.	Natural Gas Wage	Res. Build Emp.	Res. Build Wage
Post × Water-adjacent × Treated	0.039***	0.022*	0.077*	0.012	0.581*	0.153**	-0.032*	0.002
	(0.013)	(0.011)	(0.040)	(0.016)	(0.335)	(0.070)	(0.019)	(0.010)
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	651,600	747,344	138,739	138,739	25,242	25,242	901,476	832,772
R^2	0.987	0.801	0.983	0.868	0.970	0.846	0.976	0.838

Notes: This table reports estimates from fixed effects regressions examining the effect of data center exposure on sectoral employment and wage outcomes. The key coefficient of interest, Post × Water-adjacent × Treated, captures the triple-difference interaction between treated states, the post-policy period, and counties located near large water bodies. Standard errors clustered at the state level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8
Data Center Subsidies and Employment by Industry Resource Intensity

	Log Employment
	(1)
Post × Treated State × Water-adjacent × Resource-Intensive	−0.024*
	(0.017)
Post × Treated State × Water-adjacent	−0.002
	(0.008)
Post × Water-adjacent × Resource-Intensive	−0.017***
	(0.006)
Treated State × Water-adjacent × Resource-Intensive	0.003
	(0.023)
Water-adjacent × Resource-Intensive	−0.174***
	(0.029)
County × Cohort fixed effects	Yes
State × Quarter × Cohort fixed effects	Yes
Water-adjacent × Quarter × Cohort fixed effects	Yes
Industry × State fixed effects	Yes
Observations	21,381,620
R^2	0.881

Notes: The dependent variable is the natural logarithm of employment. Resource-intensive is an indicator equal to one for industries classified as resource-intensive. The coefficient on Post × Treated State × Water-adjacent gives the estimated effect for non-resource-intensive industries. The four-way interaction reports the additional effect for resource-intensive industries; therefore, the total effect for resource-intensive industries is the sum of the three-way and four-way coefficients. Lower-order terms that are absorbed by the fixed effects are not reported. Robust standard errors clustered by state are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 9
Effect of Data Center Exposure on Electricity Rates and Land Prices

	Residential electricity	Developed land	Agricultural land
Post × Water-adjacent × Treated	0.002	0.198**	-0.008
	(0.004)	(0.045)	(0.037)
County FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes
Observations	245,883	44,198	62,359
R^2	0.920	0.779	0.773

Notes: This table reports coefficients from annual stacked county-panel OLS regressions of log residential electricity rates, log developed-land prices, and log agricultural-land prices on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the post-adoption period (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-adjacent-by-year fixed effects. All fixed effects are interacted with cohort indicators. Each cohort's sample spans five years before through five years after adoption. Comparison states are never treated or not yet treated at cohort adoption. The table reports full-sample estimates. Electricity rate data use county-year observations from 2001–2024 and are from the U.S. Energy Information Administration (EIA). Land price data use county-year observations from 2010–2025 and are from Acres.com. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 10
Data Center Tax Incentives, Rents, and Home Values

	Log median gross rent (1)	Log median home value (2)
Post × Treated × Water-adjacent	-0.0079**	-0.0095*
	(0.0038)	(0.0053)
County FE	Yes	Yes
State × Year FE	Yes	Yes
Water-adjacent × Year FE	Yes	Yes
Observations	205,802	205,734
R^2	0.9562	0.9854

Notes: This table reports coefficients from annual stacked county-panel OLS regressions of log median monthly gross rent and log median owner-occupied home value on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the adoption year and subsequent years (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-adjacent-by-year fixed effects. All fixed effects are interacted with cohort indicators. Each cohort's sample spans five years before through five years after adoption, with Post equal to one beginning in the year of treatment. Comparison states are never treated or not yet treated at cohort adoption. The data use county-year observations from the American Community Survey five-year releases from 2009–2022. Dollar amounts are nominal. Standard errors clustered at the state level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 11
Effect of Data Center Exposure on Electricity Rates, Land Prices, Employment, and Political Outcomes by Period

	Residential electricity	Developed land	Agricultural land	Log employment	Incumbent lost
Panel A: Pre-2016					
Post × Water-adjacent × Treated	-0.002	0.473*	-0.004	0.003	-0.042**
	(0.006)	(0.253)	(0.094)	(0.004)	(0.018)
County FE	Yes	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes	Yes
Observations	175,921	12,504	29,942	617,866	85,605
R^2	0.901	0.821	0.830	0.999	0.15
Panel B: Post-2016					
Post × Water-adjacent × Treated	0.008*	0.355**	-0.213**	-0.013*	0.076***
	(0.004)	(0.175)	(0.100)	(0.006)	(0.028)
County FE	Yes	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes	Yes
Observations	68,419	20,863	31,716	1,177,008	24,740
R^2	0.942	0.767	0.801	0.998	0.220

Notes: This table reports estimates separately for observations before 2016 and from 2016 onward. The outcomes are the log residential electricity rate, log developed-land price, log agricultural-land price, log employment, and an indicator for whether the incumbent lost. Standard errors clustered by state are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

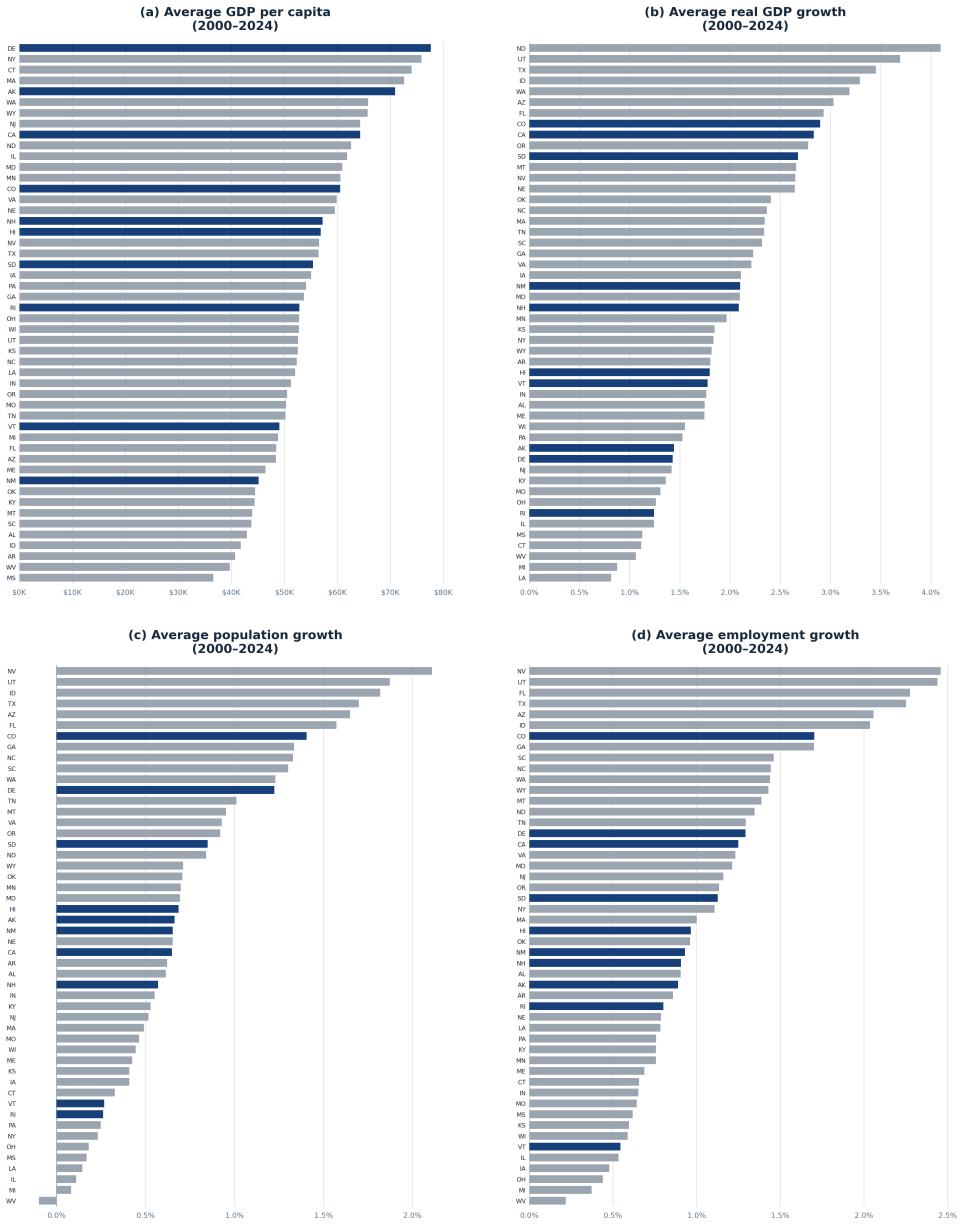
Table 12
Robustness of Data Center, Fiscal, and Employment Estimates

	(1)	(2)	(3)	(4)	(5)
Panel A: Excluding states with evidence of lobbying	Normalized data center count	Total revenue	Total taxes	Total expenditure	Log employment
Post × Treated × Water-adjacent	0.3724**	0.0240*	0.0332**	0.0334*	-0.0148**
	(0.1627)	(0.0137)	(0.0163)	(0.0180)	(0.0057)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Time FE	Yes	Yes	Yes	Yes	Yes
R^2 / Pseudo R^2	0.807	0.993	0.991	0.992	0.998
Observations	195,373	180,335	180,406	180,406	934,012
Panel B: Excluding states with incentives enacted in broader legislative packages	Normalized data center count	Total revenue	Total taxes	Total expenditure	Log employment
Post × Treated × Water-adjacent	0.2234*	0.0237*	0.0232	0.0459***	-0.0106*
	(0.1114)	(0.0130)	(0.0174)	(0.0178)	(0.0061)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Time FE	Yes	Yes	Yes	Yes	Yes
R^2 / Pseudo R^2	0.849	0.990	0.988	0.990	0.997
Observations	203,776	188,653	188,758	188,758	1,014,253
Panel C: Dillon's Rule states	Normalized data center count	Total revenue	Total taxes	Total expenditure	Log employment
Post × Treated × Water-adjacent	0.3771**	0.0351**	0.0483***	0.0515***	0.0038
	(0.1711)	(0.0163)	(0.0167)	(0.0174)	(0.0084)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Time FE	Yes	Yes	Yes	Yes	Yes
R^2 / Pseudo R^2	0.915	0.992	0.992	0.992	0.998
Observations	69,647	63,592	63,652	63,652	362,651
State clusters	14	14	14	14	14

Notes: Each panel reports separate stacked triple-difference regressions. Column 1 uses annual OLS through 2025; the data center count is divided by its 1996–2025 county–year mean (0.32521), and event year zero is the year after adoption. Columns 2–4 use annual PPML through 2022, with adoption as event year zero. Column 5 retains the quarterly OLS specification for the natural logarithm of average private employment. Time fixed effects refer to years in Columns 1–4 and quarters in Column 5; all fixed effects are interacted with cohort. All robustness samples exclude California. Standard errors are clustered by state. Panel A excludes VA, TX, OH, GA, IL, IN, MI, NC, WI, SC, and CT. Panel B excludes FL, IL, KY, MA, MN, NJ, NY, NC, OH, OR, PA, SC, and WI. Panel C restricts the sample to AZ, DE, ID, NE, NH, NJ, NC, OK, SD, TX, VT, VA, WV, and WY, using NACo's county-authority classifications; Utah is excluded because its Hutchinson's Rule grants broader local authority. The classification is fixed rather than reconstructed annually. Parentheses report standard errors. Fit statistics are R^2 for OLS and pseudo R^2 for PPML. Observations count stacked county–years in Columns 1–4 and county–quarters in Column 5. $^{*}p < 0.10$, $^{**}p < 0.05$,and $^{***}p < 0.01$.

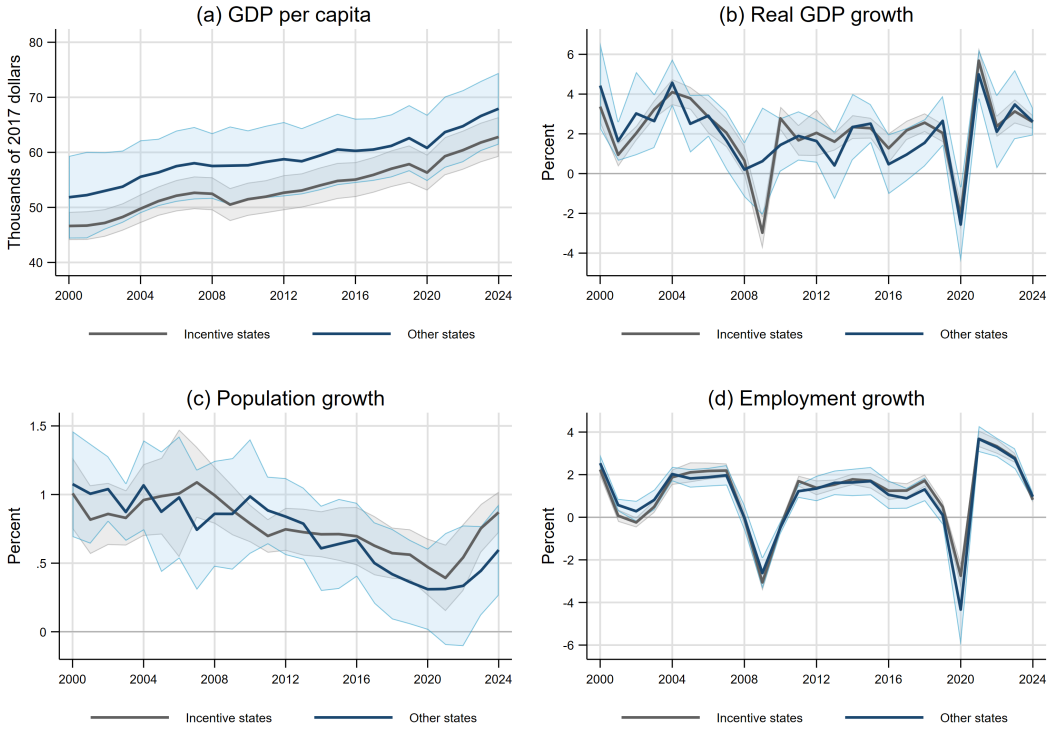
A Additional Figures

Figure A.1. Distribution of Data Center Incentive States.



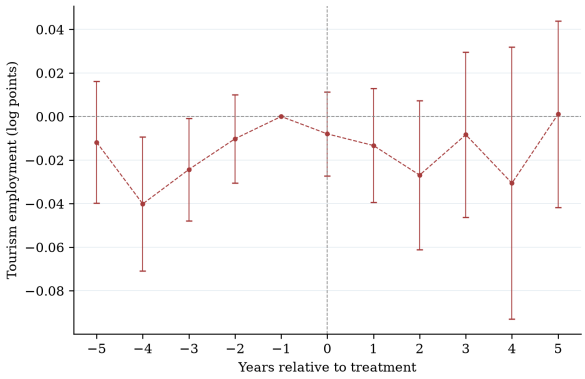
*Notes:*This figure ranks states according to their average economic characteristics between 2000 and 2024. Panels (a) through (d) present average GDP per capita, real GDP growth, population growth, and employment growth, respectively. Gray bars represent states that adopted a data center tax incentive, while dark-blue bars represent states that have not. GDP per capita is reported in thousands of 2017 dollars and growth rates are reported in percentage terms. The District of Columbia is excluded from the sample.

Figure A.2. State-Level Economic Trends

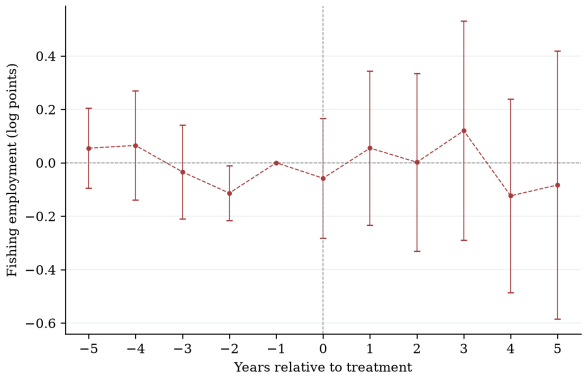


Notes: This figure compares economic trends between between 2000 and 2024 in states that adopted data center tax incentives with trends in states that never adopted a data center tax incentive. Panels (a) through (d) present average GDP per capita, real GDP growth, population growth, and employment growth, respectively. Gray lines represent incentive states, while dark-blue lines represent never treated states. Shaded areas report 95% confidence intervals calculated from cross-state variation. The District of Columbia is excluded from the sample.

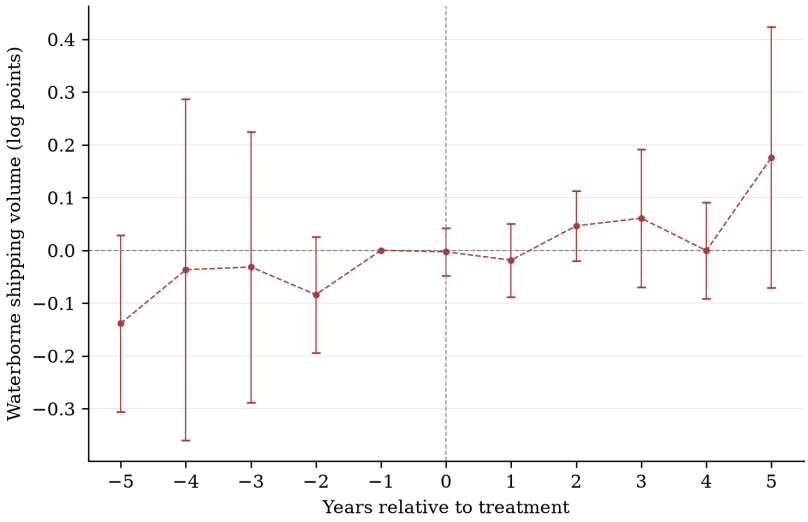
Figure A.3. Placebo Tests for Wate Related Economic Activity



(a) Tourism employment



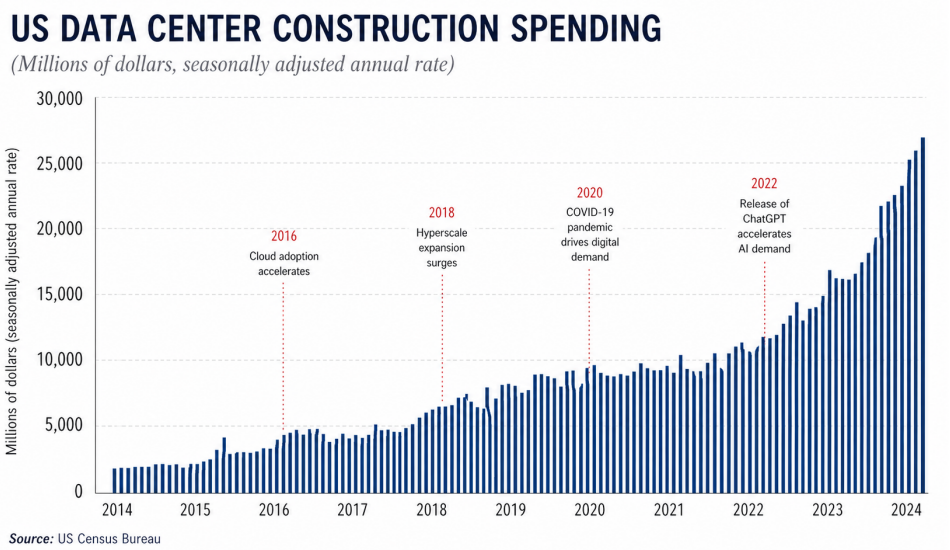
(b) Fishing employment



(c) State waterborne shipping volume

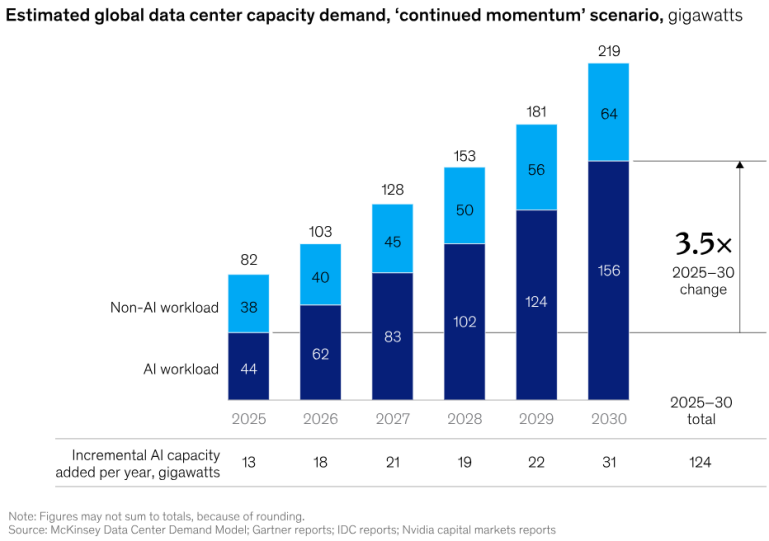
Notes: This Figure reports coefficients from a stacked county-panel regressions of water related economic activity on the interaction between an indicator for whether the state passed a data center subsidy(Treated) and an indicator for years equal or after the passage of the subsidy(Post) and an indicator for whether the county is adjacent to a water body (Water-Adjacent), with county-by-cohort and year-by-cohort fixed effects. The sample includes county-year observations from 1996-2024. Standard errors, clustered at the state level. All outcomes are expressed in natural logarithms. Event year −1 is omitted in every panel. Vertical bars report 95 percent confidence intervals, and standard errors are clustered by state.

Figure A.4. Data Center Construction Spending



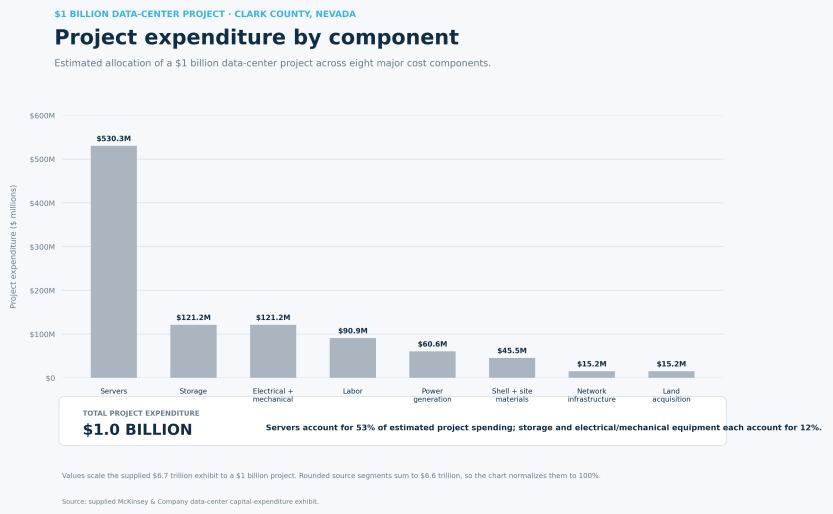
Notes: This figure reports U.S. data center construction spending from 2014 through 2024, measured in millions of dollars at a seasonally adjusted annual rate. The annotations highlight developments in cloud adoption in 2016, hyperscale expansion in 2018, digital demand during the COVID-19 pandemic in 2020, and the release of ChatGPT in 2022. Data is from the U.S. Census Bureau.

Figure A.5. Data Center Projected Growth

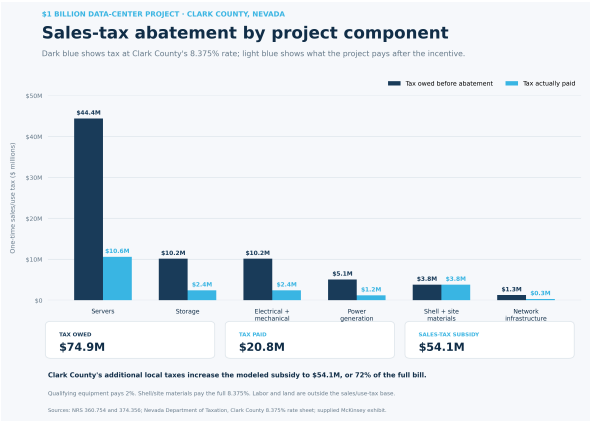


Notes: This figure reports projected global demand for data center capacity from 2025 through 2030 under McKinsey's "continued momentum" scenario. Capacity is measured in gigawatts, with dark blue indicating AI workloads and light blue indicating non-AI workloads. Total capacity demand is projected to increase from 82 gigawatts in 2025 to 219 gigawatts in 2030, while capacity demand for AI workloads increases from 44 to 156 gigawatts. The figures below the bars report incremental AI capacity added each year. Projections are from the McKinsey Data Center Demand Model.

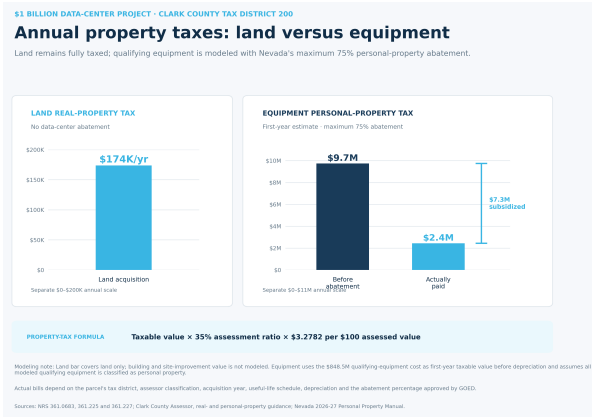
Figure A.6. Example of a Data Center Tax Incentive in Clark County, Nevada.



(a) Project expenditures by component



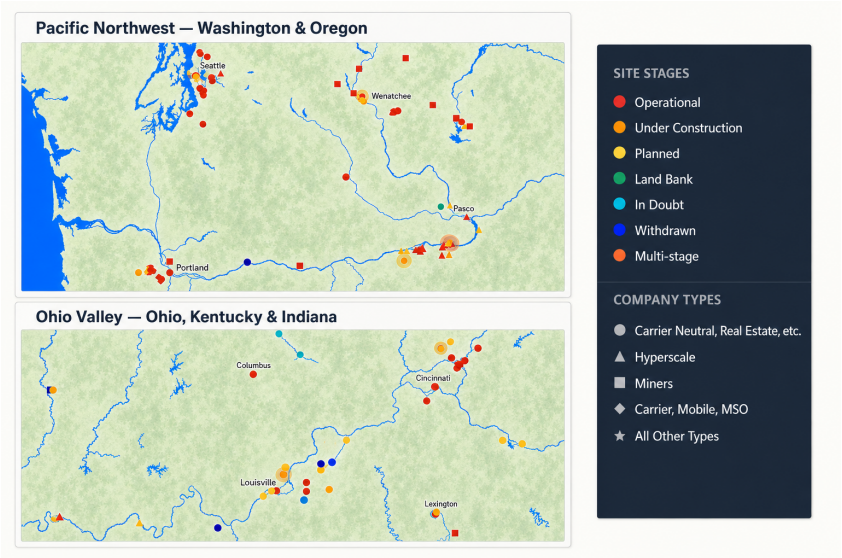
(b) Sales and use tax by component



(c) Real and personal property taxes

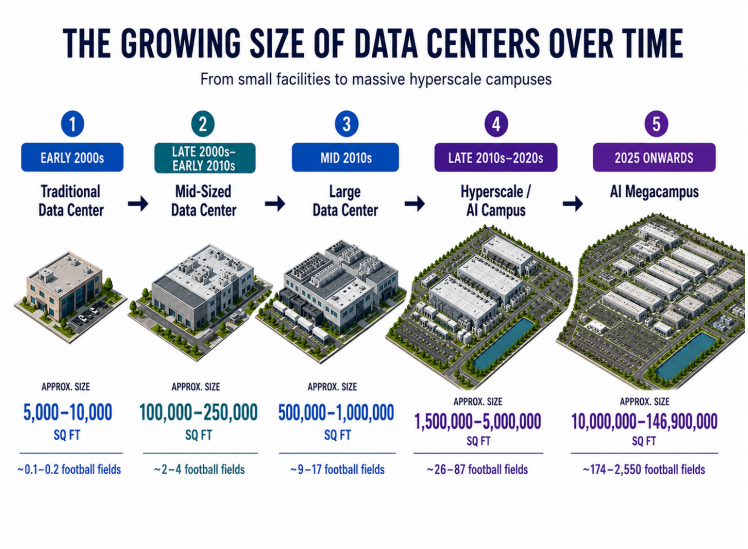
Notes: This figure illustrates the tax treatment of a hypothetical data center project in Clark County, Nevada. Panel (a) presents project expenditures by component. Panel (b) compares the sales and use taxes owed under the standard tax rate with the amount paid after the abatement. Panel (c) presents the corresponding real and personal property-tax liabilities.

Figure A.7. Data center locations and proximity to water bodies



Notes: This figure shows data center locations and nearby water bodies in the Pacific Northwest, covering Washington and Oregon (upper panel), and the Ohio Valley, covering Ohio, Kentucky, and Indiana (lower panel). Marker colors indicate project stages, while marker shapes distinguish company types, as shown in the legend. Water bodies are shown in blue.

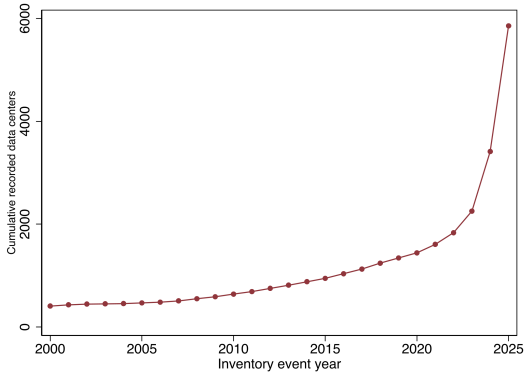
Figure A.8. Data Centers Over Time



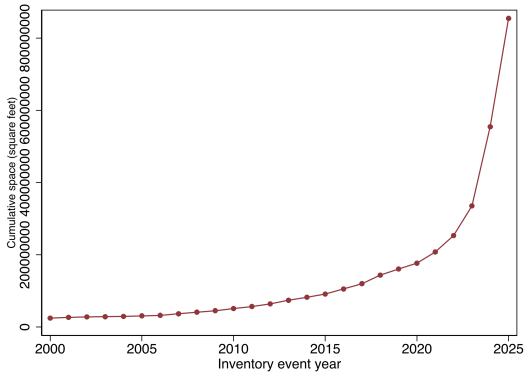
Notes: This figure illustrates the evolution of data center scale from traditional facilities in the early 2000s to hyperscale and artificial intelligence campuses. The timeline shows facility and campus types and approximate sizes reported in square feet and expressed as equivalent numbers of football fields.

Figure A.9. Growth in Data Center Scale Over Time

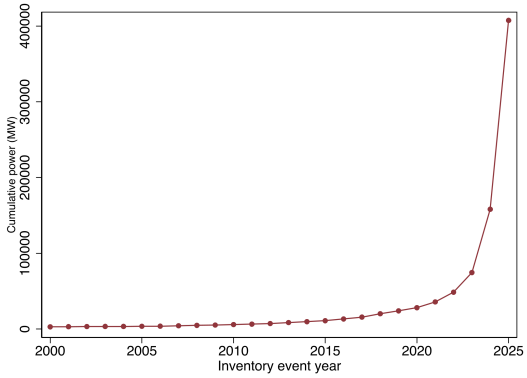
(a) Cumulative Recorded Data Centers



(b) Total Data Center Space

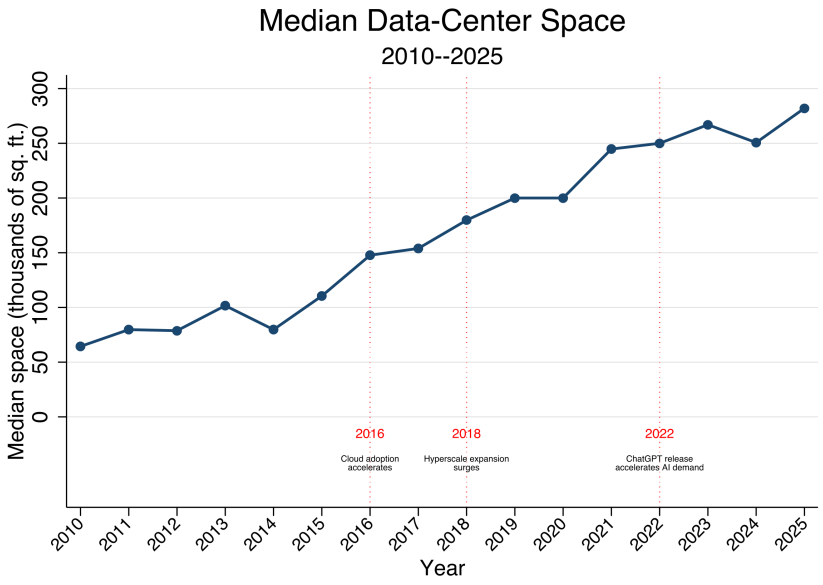


(c) Estimated Power Capacity

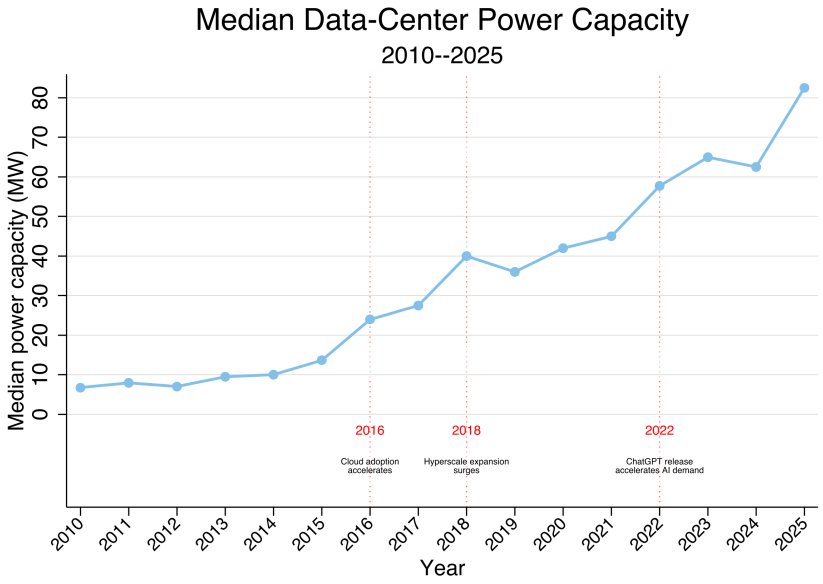


Notes: This figure plots the growth of data center activity from 2000 to 2025. Panel A reports cumulative recorded data center development events, Panel B reports cumulative data center space in square feet, and Panel C reports estimated cumulative power capacity in megawatts. The figures show that recent data center growth reflects not only an increase in the number of facilities, but also a sharp increase in facility scale and power intensity.

Figure A.10. New Data Center Scale and Power Capacity



(a) Median Data Center Space



(b) Median Data Center Power Capacity

Notes: This figure presents annual median data center size and power capacity of new data centers from 2010 through 2025. Panel (a) reports median data center space in thousands of square feet. Panel (b) reports median selected power capacity in megawatts. Red dotted lines identify important periods in the evolution of the data center industry: the acceleration of cloud adoption in 2016, the expansion of hyperscale data centers in 2018, and the release of ChatGPT and subsequent acceleration of AI-related demand in 2022. Annual medians are calculated across observations with nonmissing values.

B Additional Tables

Table B.1
Description of Key Variables

This table reports the definitions and sources of the key variables used in the analysis.

Variable	Description	Source
Number of Data Centers	Cumulative number of recorded data center projects in county c by year t. A project enters in its announcement year, or, if unavailable, construction-start, construction-finish, then activation year, in that order. Counts include announced and under-construction facilities, not only operating facilities.	Aterio; Baxtel.com
Data Center Space	Cumulative recorded facility square footage in the county, using the same project dates as the count measure.	Aterio; Baxtel.com
Data Center Power Capacity	Cumulative selected power capacity in megawatts (MW), using the same project dates as the count measure.	Aterio; Baxtel.com
Normalized Data Center Outcomes	Each cumulative outcome is divided by its mean over unique county–years in 1996–2025: 0.32521 facilities, 38,760.144 square feet, or 9.673 MW. The denominators include zero outcomes and are fixed across robustness checks. The space regression excludes observations with positive counts but zero space.	Author calculations
Treated State	Within each stacked cohort, an indicator for states belonging to the focal incentive-adoption cohort.	State statutes
Post	For the data center regressions, equals one from the calendar year after adoption onward; event year 0 is that first post year. For the revenue regressions, equals one in the adoption year and subsequent years; event year 0 is adoption. Both baseline annual windows span event years -5 through $+5$.	State statutes; author calculations
Water-adjacent County	Indicator for the Water-adjacent county classification used in the regressions.	Geographic data; author classification
Water-adjacent Neighbor	Indicator for a county that borders a waterfront county but is not itself classified as waterfront.	U.S. Census Bureau
Population	Annual estimate of the total resident population of county c in year t.	U.S. Census Bureau

Continued on next page

Table B.1 continued

Variable	Description	Source
Revenue Measurement	Financial outcomes are annual county-level aggregates in thousands of nominal dollars. The revenue table uses available observations for each outcome in 1995–2022. Reported zeros are retained; general charges and property taxes are filled with zero when missing and total revenue is reported, and missing sales-tax components are filled with zero when total taxes are reported.	U.S. Census Bureau; author calculations
Total Revenue	Constructed sum of total taxes, general charges including liquor-store revenue, and intergovernmental revenue. This measure excludes separately recorded utility revenue and miscellaneous general revenue; it is not the broader Census revenue aggregate.	U.S. Census Bureau
Total Taxes	All reported tax receipts (Census item codes T*), including property, sales, income, license, and other taxes.	U.S. Census Bureau
General Charges	Receipts from government services, plus liquor-store revenue under the study definition. Public utility receipts are recorded separately. This differs from the official general-charges aggregate because liquor-store revenue is included.	U.S. Census Bureau
Intergovernmental Revenue	Transfers received from federal, state, and other local governments (B*, C*, and D*).	U.S. Census Bureau
Sales Taxes	General sales taxes plus other selective sales taxes (T09 + T19). This is narrower than all sales and gross-receipts taxes and does not include every selective sales-tax category.	U.S. Census Bureau
Property Taxes	Property-tax receipts (T01).	U.S. Census Bureau
Public Utility Taxes	Tax receipts classified as public utility taxes (T15). These differ from receipts earned by publicly operated utilities and from utility-license taxes.	U.S. Census Bureau
License Taxes	License-tax receipts across reported license categories (T20, T21, T22, T23, T24, T25, T27, T28, and T29).	U.S. Census Bureau
Utility Revenue	Receipts from publicly operated water, electricity, gas, and transit systems (A91–A94), excluding intergovernmental transfers.	U.S. Census Bureau
Parks and Recreation Charges	Charges for parks and recreation services (A61), excluding intergovernmental transfers. These are service receipts, not all funding allocated to parks.	U.S. Census Bureau
Education Charges	Identified education-service charges (A09, A10, A12, A16, A18, and A21), excluding intergovernmental transfers and general tax funding.	U.S. Census Bureau
Sewerage Charges	Charges for sewerage services (A80), excluding intergovernmental transfers.	U.S. Census Bureau
Total Expenditure	Direct expenditure plus intergovernmental expenditure, excluding insurance-trust benefits. Includes current operations, combined capital outlays, interest, assistance and subsidies, and transfers to other governments.	U.S. Census Bureau

Continued on next page

Table B.1 continued

Variable	Description	Source
Direct Expenditure	Expenditure excluding intergovernmental transfers and insurance-trust benefits. Interest is included once.	U.S. Census Bureau
Current Operations	Routine operating expenditure, including employee compensation, supplies, and purchased services.	U.S. Census Bureau
Combined Capital Expenditure	Construction plus other capital outlays. Historical capital totals are retained through 2011; F and G items are combined in 2012–2021; F items represent combined capital from 2022.	U.S. Census Bureau
Construction Expenditure	Separately reported construction expenditure through 2021. In the overall construction regression, missing amounts are set to zero when current operations are observed and year is at least 2021; this is an analytical imputation, not separately observed construction after 2021.	U.S. Census Bureau
Other Capital Outlays	Capital expenditure excluding construction, including equipment and land purchases. Historical capital totals less construction through 2011; G items in 2012–2021. Not separately available from 2022.	U.S. Census Bureau
Interest on Debt	Interest payments on debt. Excludes principal repayments and debt outstanding.	U.S. Census Bureau
Assistance and Subsidies	Reported assistance and subsidy expenditure in harmonized J items. Coverage is more limited than for aggregate expenditure.	U.S. Census Bureau
Intergovernmental Expenditure	Transfers paid to other governments. Adding transfers to direct expenditure gives gross expenditure, not a consolidated total net of transfers across governments.	U.S. Census Bureau
Utilities Expenditure	Direct spending on publicly operated water, electricity, gas, and transit systems. Excludes transfers.	U.S. Census Bureau
Infrastructure Expenditure	Direct spending on highways, airports, ports, parking, sewerage, and solid waste. Utilities are separate. Excludes transfers.	U.S. Census Bureau
Services Expenditure	Direct spending on police, fire, parks, libraries, education, health, hospitals, welfare, and housing/community development. Excludes transfers.	U.S. Census Bureau
Total Education Expenditure	Education expenditure including current operations, capital expenditure, applicable other direct spending, and intergovernmental transfers.	U.S. Census Bureau
Education Construction Expenditure	Separately reported construction spending on education facilities, available through 2021. Missing values are not filled using the overall construction rule.	U.S. Census Bureau
Reading Scores	All-student reading achievement in cohort-scale standard-deviation units. District scores are aggregated to county–years using subject-specific assessment counts, among geographically matched single-county districts. Available years are 2009–2025, excluding 2020–2021.	Stanford Education Data Archive (SEDA 2025.2)

Continued on next page

Table B.1 continued

Variable	Description	Source
Math Scores	All-student math achievement in cohort-scale standard-deviation units. District scores are aggregated to county–years using subject-specific assessment counts, among geographically matched single-county districts. Available years are 2009–2025, excluding 2020–2021.	Stanford Education Data Archive (SEDA 2025.2)
Average Yield	Volume-weighted average yield for a CUSIP in a given month. Volume refers to the par value of the trade.	MSRB
Yield Spread	Calculated as the difference between the *Average Yield* and the coupon-equivalent risk-free yield. The risk-free yield is calculated using the present value of the bond's coupon payments and face value, discounted using the U.S. Treasury yield curve and maturity-matched zero-coupon yields from Gürkaynak et al. (2007). The yield spread is calculated following Longstaff et al. (2005): $\text{Yield Spread}_{i,t} = \text{Average Yield}_{i,t} - \text{Risk-Free Yield}_{i,t}$.	MSRB, FEDS

Table B.2
State Data Center Tax Incentives

State	Year	State Sales	Local Sales	Local Property Taxes	Investment Threshold	Jobs	Wage Requirement
AL	2012	Yes	Yes	Yes	$200M; $400M	20	$40K+
AZ	2013	Yes	Yes	No	$25M; $50M	None	None
AR	2023	Yes	Yes	No	$500M	25	None
CT	2021	Yes	None	No	$200M	30	None
FL	2017	Yes	Yes	No	$150M	20	None
GA	2018	Yes	Yes	No	$250M	25	None
ID	2020	Yes	None	No	$250M	30	None
IL	2019	Yes	Yes	No	$250M	20	120% of median wage
IN	2019	Yes	None	No	$25M; $150M	None	None
IA	2009	Yes	Yes	No	$200M; $1B	10; 50	None
KS	2025	Yes	Yes	No	$250M	20	None
KY	2023	Yes	None	No	$100M; $450M	None	None
LA	2024	Yes	Yes	Yes	$200M	50	None
ME	2021	Yes	None	No	None	20	None
MD	2020	Yes	None	No	$2M	5	None
MA	2024	Yes	None	No	$50M	100	None
MI	2016	Yes	None	No	$250M	30	150% of median wage
MN	2012	Yes	None	Yes	$30M	None	None
MS	2010	Yes	No	No	$20M	20	150% of mean wage
MO	2015	Yes	Yes	No	$25M	10	None
MT	2017	None	None	Yes	$50M	None	None
NE	2012	Yes	Yes	Yes	None	None	None
NV	2015	Yes	Yes	Yes	$25M; $100M	10; 50	None
NJ	2024	Yes	None	No	None	None	None
NY	2000	Yes	Yes	No	None	None	None
NC	2016	Yes	Yes	No	$75M	None	None
ND	2021	Yes	Yes	No	$150M	None	None
OH	2011	Yes	Yes	No	$100M; $500M	10; 50	None
OK	2010	Yes	Yes	No	$5M	5	125% of mean wage
OR	2015	None	None	Yes	$50M	10–35	None
PA	2022	Yes	Yes	No	$75M	25	None
SC	2012	Yes	Yes	No	$50M	None	None
TN	2016	Yes	Yes	No	$100M	15	None
TX	2013	Yes	Yes	No	$200M	20	120% of mean wage
UT	2016	Yes	Yes	No	$150M	15	110% of mean wage
VA	2009	Yes	Yes	No	$150M; $1B	50; 25	None
WA	2010	Yes	Yes	No	$25M	20	None
WI	2023	Yes	Yes	No	$150M	30	150% of mean wage
WV	2016	Yes	None	Yes	$50M	50	None
WY	2010	Yes	Yes	No	$50M	20	None

Notes: This table summarizes state-level data center incentive provisions, including the year of adoption, whether the incentive applies to state sales taxes, local sales taxes, and local property taxes, and the associated investment, employment, and wage requirements. Dollar values are reported in millions unless otherwise indicated.

Table B.3
Water adjacent vs Non-water adjacent Observable characteristics in Year 2000

Characteristic	Waterfront difference	t-statistic	p-value
Panel A: Percentage-point measures			
Minority share	0.575	1.674	0.101
Female share	0.051	0.475	0.637
Poverty rate	−0.526	−1.758	0.085
Unemployment rate	0.061	0.417	0.679
Information-sector employment share	0.053	1.635	0.109
Panel B: Logged characteristics			
Log median age	−0.006	−0.994	0.325
Log mean commute time	0.005	0.482	0.632

Notes: Each row compares waterfront and non-waterfront counties using year 2000 data. Positive differences indicate higher values in waterfront counties. Panel A is measured in percentage points. Commute time is measured in minutes.

Table B.4
Summary of State Data Center Tax Incentives

Measure	Statistic	All	Early adopters through 2016	Late adopters after 2016
Incentives adopted	Number of states	40	22	18
	Percent of adopters	100.0%	55.0%	45.0%
State sales-tax abatement	Number of states	40	22	18
	Percent of states	100.0%	100.0%	100.0%
Local sales-tax abatement	Number of states	39	21	18
	Percent of states	97.5%	95.5%	100.0%
Property-tax abatement	Number of states	8	6	2
	Percent of states	20.0%	27.3%	11.1%
Investment threshold	Number of states	36	20	16
	Percent of states	90.0%	90.9%	88.9%
	Mean threshold ($M)	$110.8	$80.9	$147.3
Employment threshold	Number of states	29	16	13
	Percent of states	72.5%	72.7%	72.2%
	Mean threshold (jobs)	17.3	13.2	22.2
Wage threshold	Number of states	8	6	2
	Percent of states	20.0%	27.3%	11.1%

Notes: Early adopters include states that adopted an incentive through 2016.

Table B.5
Data Center Growth Over Time

Year	Total DCs	New DCs	Total Space (sq. ft.)	New Space (sq. ft.)	Total Power (MW)	New Power (MW)	Median DC Space (sq. ft.)	Median Power (MW)
2000	408	406	24,697,480	24,283,881	2,885	2,847	33,000	2.195
2001	433	25	26,724,056	2,026,574	3,070	185	86,000	7.000
2002	447	14	28,046,290	1,322,235	3,261	191	48,627	4.110
2003	452	5	28,574,400	528,110	3,372	111	433,333	35.000
2004	457	5	29,419,076	844,675	3,429	57	40,000	7.000
2005	469	12	30,926,318	1,507,243	3,609	180	34,000	8.750
2006	483	14	32,226,518	1,300,200	3,759	150	85,440	10.000
2007	509	26	36,573,572	4,347,053	4,322	564	134,550	12.250
2008	550	41	41,155,872	4,582,302	4,820	498	77,300	6.000
2009	589	39	45,276,540	4,120,666	5,235	415	84,000	8.000
2010	640	51	51,163,256	5,886,719	5,883	648	64,500	6.720
2011	689	49	56,852,296	5,689,037	6,454	571	79,914	7.965
2012	752	63	64,089,392	7,237,097	7,271	818	78,760	7.000
2013	815	63	73,983,872	9,894,482	8,543	1,271	101,800	9.500
2014	882	67	82,544,432	8,560,557	9,692	1,150	80,000	10.000
2015	946	64	91,267,848	8,723,417	10,955	1,262	110,620	13.680
2016	1,036	90	105,221,632	13,953,786	13,266	2,312	147,950	24.000
2017	1,127	91	120,098,696	14,877,061	15,701	2,435	154,000	27.500
2018	1,241	114	143,640,528	23,541,834	20,214	4,513	180,000	40.000
2019	1,343	102	160,724,992	17,084,459	23,994	3,779	200,000	35.985
2020	1,442	99	176,718,304	15,993,310	28,126	4,132	200,000	42.000
2021	1,608	166	207,758,096	31,039,805	35,786	7,660	245,000	45.000
2022	1,833	225	253,243,440	45,485,332	48,582	12,796	250,000	57.760
2023	2,253	420	335,423,040	82,179,605	74,575	25,993	267,000	65.000
2024	3,413	1,160	554,545,088	219,122,047	158,137	83,562	250,750	62.505
2025	5,855	2,442	854,548,160	300,003,049	407,402	249,265	282,000	82.500

Notes: Cumulative counts, facility square footage, and power capacity are summed across counties in the geographic panel. Inventory event dates prioritize announcement, construction start, construction finish, and activation; these are not activation-only counts.

Table B.6
Data Center Size and Power Capacity by Inventory Event Period

Event period	*N*	Mean	Median	SD	P25	P75
Panel A: Facility size (square feet)						
Before 2016	842	108,787	58,098	148,135	23,500	138,000
2016 or later	2,831	269,933	250,000	159,869	180,000	305,200
All years	3,673	232,992	222,000	171,210	124,780	288,000
Panel B: Selected power capacity (MW)						
Before 2016	911	12.04	5.00	17.79	1.49	15.00
2016 or later	4,605	86.09	66.00	71.32	48.00	100.00
All years	5,516	73.86	60.00	71.09	34.00	90.00

Notes: The table reports facilities with selected inventory event dates between 2000 and 2025. The event date prioritizes announcement, then construction start, construction finish, and activation. The post-2016 group includes event years 2016 or later. Space is in square feet and power in MW. SD denotes standard deviation; P25 and P75 are the 25th and 75th percentiles.

Table B.7
Facility Size and Power Capacity Before and After 2016

	(1) Log facility size (square feet)	(2) Log power capacity (MW)
Inventory event in 2016 or later	1.337***	2.399***
	(0.109)	(0.166)
State fixed effects	Yes	Yes
Observations	3,668	5,514
R^2	0.376	0.538

Notes: The sample covers selected inventory event dates from 2000 through 2025, prioritizing announcement, construction start, construction finish, and activation. The dependent variables are log facility square footage and log selected power capacity. The post-2016 indicator equals one for event years 2016 or later. Both models include state fixed effects and state-clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.8
Data Center Tax Incentives, Population, and Migration

	(1)	(2)
Panel A: Through 2022	Log population	Net migration per 1,000
Post × Treated × Waterfront	-0.0023	0.2587
	(0.0024)	(0.4597)
p-value	0.353	0.576
County FE	Yes	Yes
State × Year FE	Yes	Yes
Waterfront × Year FE	Yes	Yes
R^2	0.9996	0.4216
Observations	265,633	247,299

Notes: This table reports annual stacked triple-difference OLS estimates. Column 1 uses the natural logarithm of county population; Column 2 uses total net migration per 1,000 residents, combining domestic and international net migration. Population is measured on July 1, and migration covers the preceding July–June period. Census migration rates use the average of beginning- and end-period population as the denominator. Population observations begin in 2000; annual migration observations begin in 2001. Event year zero is the adoption year, and the window spans −5 to +5 years. Standard errors clustered by state are in parentheses. Observations count stacked county–years. Source: U.S. Census Bureau Population Estimates Program. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table B.9
Additional Robustness Checks

	(1)	(2)	(3)	(4)	(5)
Panel A: Four years before and after event year zero	Normalized data center count	Total revenue	Total taxes	Total expenditure	Log employment
Post × Treated × Waterfront	0.3052**	0.0161*	0.0209*	0.0252**	−0.010**
	(0.1154)	(0.0095)	(0.0117)	(0.0115)	(0.004)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
R^2 / Pseudo R^2	0.879	0.993	0.992	0.993	0.998
Observations	241,715	225,067	225,182	225,182	1,211,182
Panel B: Three years before and after event year zero	Normalized data center count	Total revenue	Total taxes	Total expenditure	Log employment
Post × Treated × Waterfront	0.2886**	0.0119	0.0170*	0.0173*	−0.009**
	(0.1232)	(0.0081)	(0.0091)	(0.0090)	(0.004)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
R^2 / Pseudo R^2	0.904	0.994	0.993	0.994	0.998
Observations	189,361	177,239	177,334	177,334	932,384
Panel C: Excluding Virginia and California	Normalized data center count	Total revenue	Total taxes	Total expenditure	Log employment
Post × Treated × Waterfront	0.3043**	0.0202**	0.0261**	0.0310**	-0.0123**
	(0.1213)	(0.0103)	(0.0120)	(0.0126)	(0.0048)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
R^2 / Pseudo R^2	0.846	0.992	0.991	0.992	0.998
Observations	290,254	269,912	270,047	270,047	1,473,191

Table B.9 (continued)

Panel D: Standard errors clustered by state and year	Normalized data center count	Total revenue	Total taxes	Total expenditure	Log employment
Post × Treated × Waterfront	0.3407**	0.0203**	0.0247**	0.0308***	−0.010**
	(0.1503)	(0.0094)	(0.0115)	(0.0119)	(0.005)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
R^2 / Pseudo R^2	0.846	0.992	0.991	0.992	0.998
Observations	293,180	271,501	271,637	271,637	1,483,920
Panel E: Excluding the 10 largest counties	Normalized data center count	Total revenue	Total taxes	Total expenditure	Log employment
Post × Treated × Waterfront	0.3023***	0.0185*	0.0227*	0.0286**	−0.010**
	(0.1077)	(0.0102)	(0.0122)	(0.0117)	(0.005)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
R^2 / Pseudo R^2	0.799	0.992	0.991	0.992	0.998
Observations	292,531	270,877	271,013	271,013	1,479,824
Panel F: Excluding the 20 largest counties	Normalized data center count	Total revenue	Total taxes	Total expenditure	Log employment
Post × Treated × Waterfront	0.3210***	0.0124	0.0160	0.0261**	−0.010**
	(0.1062)	(0.0122)	(0.0123)	(0.0121)	(0.005)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
R^2 / Pseudo R^2	0.777	0.991	0.990	0.991	0.998
Observations	291,639	270,007	270,143	270,143	1,474,696

Notes: Each panel reports separate stacked triple-difference regressions. Column 1 uses annual OLS through 2025; the data center count is divided by its 1996–2025 county–year mean (0.32521), and event year zero is the year after adoption. Columns 2–4 use annual PPML through 2022, with adoption as event year zero. Column 5 retains the quarterly OLS specification for the natural logarithm of average private employment. Time fixed effects refer to years in Columns 1–4 and quarters in Column 5; all fixed effects are interacted with cohort. All robustness samples exclude California. Panels A–B shorten each model's event window to four and three years on either side of its treatment reference. Panel C also excludes Virginia in every column; Texas is retained. Standard errors are clustered by state except in Panel D, which clusters by state and year. Panels E–F retain each model's original county exclusion lists; employment exclusions use the July 1, 2025 population ranking. Parentheses report standard errors. Fit statistics are R^2 for OLS and pseudo R^2 for PPML. Observations count stacked county–years in Columns 1–4 and county–quarters in Column 5. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

C Where Data Centers Locate

Data center location decisions depend on both the type of facility and the services it provides. All data centers require electricity, cooling, land, and digital connectivity, but not every facility must remain close to its users. Location choices therefore reflect trade-offs among operating costs, proximity to customers, performance requirements, regulatory considerations, and infrastructure availability.

C.0.1 Proximity to Users

The importance of proximity varies across data center types. Third-party colocation and edge facilities are generally located relatively close to the customers they serve. Hospitals and financial institutions, for example, may rely on urban facilities because their activities require low latency, dependable access, data security, and compliance with regulatory requirements.

Cloud and hyperscale providers have greater geographic flexibility because they sell access to virtualized computing resources over the internet. Batch-processing operations, data storage, artificial-intelligence model training, and other activities that are relatively insensitive to small transmission delays can therefore be conducted in lower-cost locations farther from users. As a result, these facilities are more likely to locate where electricity, land, and cooling resources are abundant and inexpensive.

C.0.2 Electricity Availability and Cost

Electricity is one of the largest operating expenses for a data center. Firms therefore prefer locations with abundant, reliable, and relatively inexpensive power. Because data centers operate continuously, even brief service interruptions can be costly. The local grid must also have sufficient generation and transmission capacity to accommodate their substantial and persistent electricity demand.

Access to renewable energy has become another consideration as technology companies pursue sustainability commitments and attempt to reduce the carbon intensity of their operations. Nevertheless, the availability and reliability of electricity remain the primary concerns, particularly for hyperscale and artificial-intelligence facilities with exceptionally large power requirements.

C.0.3 Water and Cooling Resources

Servers produce considerable heat and must be cooled continuously to operate safely. Although data centers use several cooling technologies, many facilities rely partly on water-based systems because of their efficiency. Access to dependable water supplies may therefore make locations near rivers, lakes, reservoirs, and other water sources particularly attractive.

At the same time, data center development can place additional pressure on local water systems, especially during droughts or periods of high residential demand. Communities hosting large facilities may need to expand water-treatment, distribution, or wastewater infrastructure to accommodate this additional demand.

C.0.4 Digital Connectivity

Data centers require high-capacity internet connections and access to fiber-optic networks to transmit large volumes of information quickly and reliably. Locations near major fiber routes and internet exchange points offer lower latency and greater network redundancy.

Connectivity is particularly important for colocation and edge facilities serving hospitals, financial firms, and other customers requiring immediate access to data. Although large cloud facilities can operate farther from their users, they still require extensive fiber infrastructure to provide reliable computing and storage services.

C.0.5 Land and Physical Infrastructure

Large cloud and hyperscale data centers require substantial parcels of land for server buildings, electrical substations, cooling systems, backup generators, and security infrastructure. These facilities often locate outside dense urban areas, where land is less expensive and large industrial sites are more readily available.

Proximity to transmission lines, electrical substations, fiber networks, highways, and other supporting infrastructure can reduce construction costs and development delays. Local zoning rules and the speed of the permitting process can also affect whether a proposed location is feasible.

C.0.6 Tax Incentives and Government Support

State and local governments frequently offer property-tax abatements, sales-tax exemptions, infrastructure assistance, and expedited permitting to attract data center investment. These incentives can influence the selection among locations with otherwise similar characteristics.

However, financial incentives cannot compensate for inadequate physical infrastructure. For policymakers seeking to attract data centers, access to reliable electricity, sufficient water resources, and high-capacity internet generally matters more than tax incentives. Moreover, generous incentives may reduce the direct fiscal benefits that communities receive from hosting these facilities, even when local governments must finance additional supporting infrastructure.

C.0.7 Climate and Environmental Risk

Climate conditions affect both cooling costs and operational reliability. Cooler climates can reduce the energy required to maintain safe server temperatures. Firms also consider exposure to floods, hurricanes, wildfires, droughts, and extreme heat because interruptions to data center operations can be extremely costly.

Facilities therefore require backup power, redundant network connections, and emergency plans, particularly when locating near bodies of water or in regions exposed to natural hazards. These environmental considerations may create a tradeoff: access to water can improve cooling efficiency but may also increase exposure to flooding or competition over scarce water supplies.

Overall, data centers are not distributed randomly across locations. Urban colocation and edge facilities tend to remain close to customers with latency and compliance requirements, whereas large cloud, hyperscale, and artificial-intelligence facilities have greater flexibility to locate in lower-cost areas. Across all types, reliable electricity, water availability, high-capacity internet, and supporting infrastructure remain central to location decisions. These requirements also imply that the local consequences of data center development may be concentrated in communities that must expand or upgrade their electricity, water, and communications infrastructure.

D Types of Data Centers

Data centers differ in their ownership, scale, location, and the computing services they provide. These differences shape their infrastructure requirements, proximity to users, and effects on local electricity and water systems. Understanding the different types of data centers therefore helps explain why some facilities locate near major urban markets, while others locate in lower-cost areas with abundant land and natural resources.

D.1 Enterprise Data Centers

Enterprise data centers are owned and operated by individual organizations for their internal computing needs. Large corporations, financial institutions, hospitals, universities, and government agencies may maintain these facilities to exercise direct control over their data, security, and regulatory compliance. Enterprise data centers are often located on the organization's property or near its principal operations. However, many organizations have moved part of their computing activity to colocation centers or cloud providers.

D.2 Third-Party Colocation Centers

Third-party colocation centers provide physical space, electricity, cooling, connectivity, and security to multiple customers. Rather than constructing an independent facility, a firm rents space and installs its own servers and networking equipment inside a shared data center. The customer generally retains control over its equipment, while the colocation provider manages the building and its supporting infrastructure.

Colocation centers are frequently located near the firms they serve. Hospitals, financial institutions, and other organizations with strict latency, security, or regulatory requirements may prefer urban colocation facilities that provide rapid and dependable access to their computing systems. Because these facilities serve multiple customers, they can also provide connections to several telecommunications networks and cloud platforms from a

single location.

D.3 Cloud Data Centers

Cloud data centers are operated by providers that sell computing power, storage, databases, and software as services over the internet. Customers purchase these resources when needed rather than owning and maintaining the underlying physical equipment. Because users access cloud services remotely, cloud facilities do not always need to be located near individual customers.

This flexibility allows cloud providers to construct large facilities in areas with inexpensive land, abundant electricity, sufficient water, and strong fiber-optic connectivity. Batch-processing operations, data storage, and other activities that are relatively insensitive to small transmission delays can be performed efficiently from these lower-cost hubs.

D.4 Hyperscale Data Centers

Hyperscale data centers are exceptionally large facilities designed to support the operations of major cloud-computing and technology companies. They contain large numbers of servers and are designed to expand computing capacity rapidly as demand increases. Although many hyperscale facilities provide cloud services, the term "hyperscale" refers primarily to their size and scalability rather than to a distinct service model.

These facilities require substantial electrical capacity, large parcels of land, extensive cooling infrastructure, and high-capacity network connections. Consequently, they can have especially pronounced effects on local electricity grids, water systems, land use, and infrastructure investment.

D.5 Edge Data Centers

Edge data centers are smaller facilities located close to users or connected devices. Their purpose is to process information near the point where it is generated rather than transmit-

ting everything to a distant cloud facility. This reduces latency and supports applications requiring nearly immediate responses, including telecommunications services, autonomous systems, online gaming, and certain healthcare and financial activities.

Although an individual edge facility is smaller than a cloud or hyperscale center, a provider may operate many such facilities across different geographic markets. Their location decisions therefore emphasize proximity to users and network connectivity more heavily than access to large quantities of inexpensive land.

D.6 Artificial-Intelligence and High-Performance Computing Facilities

Artificial-intelligence and high-performance computing facilities support computationally intensive workloads, including model training, scientific simulations, and large-scale data analysis. These workloads often use specialized processors and require substantially more electricity per server than traditional computing activities. Their high equipment density also produces considerable heat, increasing the need for advanced cooling systems.

Some artificial-intelligence applications require quick responses and therefore benefit from being located near users. However, artificial-intelligence model training and many batch-processing workloads are less sensitive to distance. These activities can operate more efficiently from large, lower-cost cloud hubs with abundant power and cooling resources.

The boundaries among these categories are not always sharp. A hyperscale facility may provide cloud services and artificial-intelligence computing, while a colocation center may lease space to cloud providers as well as hospitals and financial firms. Nevertheless, these distinctions help explain the different considerations involved in data center location decisions.

E Theory: Dynamic Local Effects of Data Center Expansion

This section develops a dynamic model of how large data center projects affect local employment, resource use, and welfare. The model studies when subsidies and demand-driven expansions create net local job gains rather than reallocating employment across sectors.

The model features a scalable data center, non-data center industries that differ in resource intensity and complementarity, households that supply labor and bear local costs, and a government choosing subsidies under demand uncertainty. The main result is that employment gains are not guaranteed: when demand is high and local resources are constrained, data center growth can crowd out other industries and reduce welfare even when the project appeared beneficial ex ante.

E.1 Setup

Consider a locality populated by heterogeneous households, a data center sector indexed by D, and a collection of non-data center industries indexed by $j \in \mathcal{J}$. Time is discrete, $t = 0, 1, 2, \ldots$. The aggregate local state is given by the demand state

$$\theta_t,$$

where θ_t captures the strength of demand for data center activity (for exam- ple, AI compute demand or major shifts in the demand for cloud and related digital infrastructure).

The demand state evolves according to

$$\theta_{t+1} \sim \mathcal{P}(\cdot \mid \theta_t), \tag{E.1}$$

where $\mathcal{P}$ is a Markov transition kernel that captures how demand for data center services

changes over time.

Households can supply up to $\bar{N}_t$ units of labor in each period. Let N_t^D denote data center employment and N_t^O denote total employment in all other local industries. Total employment and unemployment are

$$N_t = N_t^D + N_t^O, \qquad U_t = \bar{N}_t - N_t \geq 0. \tag{E.2}$$

The labor supply ceiling $\bar{N}_t$ is exogenous in the short run.

E.2 Data Center Production, Profits, Subsidies, and Adjustment

The data center chooses labor n_{Dt}, resource use r_{Dt}, and investment i_{Dt}. Capital evolves according to

$$k_{D,t+1} = (1 - \delta_D)k_{Dt} + i_{Dt}, \tag{E.3}$$

where $\delta_D \in (0, 1)$ is the depreciation rate.

Data center output is determined by a standard production function

$$y_{Dt} = \tilde{F}_D(k_{Dt}, n_{Dt}, r_{Dt}), \tag{E.4}$$

where $\tilde{F}_D$ maps capital, labor, and local resource use into output. The phys- ical production technology does not depend on the demand state.

Instead, the demand state enters on the revenue side. Let the output price of data center services be

$$p_t^D = p^D(\theta_t), \tag{E.5}$$

where $p^D(\theta_t)$ is increasing in θ_t. A higher demand state therefore raises the revenue generated by a given level of output and, through that channel, increases the incentive to invest and expand.

Local policy includes a project-specific subsidy with generosity $\sigma_t \in \mathbb{R}_+$, where larger

σ_t denotes a more generous package of tax abatements, cash grants, and infrastructure support. The subsidy enters the data center's payoff via a transfer function $S_D(\sigma_t, k_{Dt}, i_{Dt})$.

Period-t profits are

$$\pi_{Dt} = p^D(\theta_t)\, y_{Dt} - w_t n_{Dt} - q_t r_{Dt} - \rho_t i_{Dt} - \Gamma_D(i_{Dt}, k_{Dt}) + S_D(\sigma_t, k_{Dt}, i_{Dt}), \tag{E.6}$$

where w_t is the local wage, q_t is the local resource price, ρ_t is the purchase price of installed capital, $\Gamma_D(i, k)$ is an adjustment cost, and $S_D(\sigma_t, k_{Dt}, i_{Dt})$ is the subsidy or policy transfer.

Assumption 1 (Data center technology, demand, generosity, and adjustment).

1. $\tilde{F}_D$ is increasing and concave in capital k_{Dt}, labor n_{Dt}, and resources r_{Dt}.
2. $p^D(\theta)$ is increasing in θ.
3. $\Gamma_D(i, k)$ is nonnegative, convex in i, continuously differentiable, and satisfies $\Gamma_D(0, k) = 0$.
4. $S_D(\sigma_t, k, i)$ is increasing in policy generosity σ_t, so that more generous subsidies raise the effective return to installed capital and investment.

Convex adjustment costs imply that changing scale rapidly is costly, which delivers gradual rather than instantaneous expansion paths. Because the demand state raises the effective output price $p^D(\theta_t)$, higher θ_t increases the marginal value of expanding data center capacity. Throughout the model, θ_t affects local outcomes through the value of data center output and the induced expansion path for k_{Dt}, not through the underlying physical produc- tion technology.

E.3 Industry Heterogeneity and Resource Use

Each non-data center industry $j \in \mathcal{J}$ is characterized by two reduced-form parameters:

- resource intensity $\alpha_j^R \geq 0$, which measures how strongly the industry's employment is exposed to local resource prices;

- complementarity $\alpha_j^C \geq 0$, which measures how strongly the industry's employment benefits from the presence or scale of data center activity.

Industry j uses capital K_{jt}, labor N_{jt}, and local resources R_{jt} to produce

$$Y_{jt} = F_j(K_{jt}, N_{jt}, R_{jt}), \tag{E.7}$$

where F_j is increasing and concave in its inputs.

Industry j solves

$$\max_{K_{jt},\, N_{jt},\, R_{jt}} \left\{ p_{jt} Y_{jt} - r_t K_{jt} - w_t N_{jt} - q_t R_{jt} \right\}, \tag{E.8}$$

which yields labor and resource demand schedules

$$N_{jt} = N_j(w_t, r_t, q_t, k_{Dt}; \alpha_j^R, \alpha_j^C), \qquad R_{jt} = R_j(w_t, r_t, q_t; \alpha_j^R, \alpha_j^C).$$

Dependence on k_{Dt} in labor demand is reduced-form and captures demand and productivity linkages whereby larger data center scale raises activity in complementary industries, beyond the indirect effect operating through local resource prices.

Aggregate employment and resource use outside the data center sector are

$$N_t^O = \sum_{j \in \mathcal{J}} N_{jt}, \qquad R_t^O = \sum_{j \in \mathcal{J}} R_{jt}. \tag{E.9}$$

Let R_{ht} denote household h's use of local resources (electricity, water, land, and related services), and define aggregate household resource use as

$$R_t^H = \sum_{h \in \mathcal{H}} R_{ht}. \tag{E.10}$$

Total local resource use is

$$R_t = r_{Dt} + R_t^O + R_t^H = r_{Dt} + \sum_{j \in \mathcal{J}} R_{jt} + \sum_{h \in \mathcal{H}} R_{ht}. \tag{E.11}$$

We collapse scarcity, congestion, and stress into a single reduced-form harm function by letting aggregate resource use R_t generate a scalar stress cost:

$$\Omega_t = \Omega(R_t), \tag{E.12}$$

where $\Omega(R_t)$ summarizes congestion, reliability pressure, environmental stress, and unpriced local harms from heavy resource use.

Assumption 2 (Industry heterogeneity and spillovers).

1. $\Omega(R)$ is increasing and convex in R.
2. $\dfrac{\partial N_j}{\partial q_t} < 0$ for all j, and $\left|\dfrac{\partial N_j}{\partial q_t}\right|$ is increasing in α_j^R.
3. $\dfrac{\partial N_j}{\partial k_{Dt}} \geq 0$ for all j, and $\dfrac{\partial N_j}{\partial k_{Dt}}$ is increasing in α_j^C.

Assumption 2 implies that all industries are weakly harmed by higher lo- cal resource prices, but highly resource-intensive industries are harmed more. At the same time, industries with stronger complementarities to data center scale benefit more from expansion in k_{Dt}.

E.4 Labor Market Closure

To permit unemployment in the simplest possible way, assume a downward- rigid wage floor $w_t \geq \bar{w}_t$. Employment is demand determined:

$$N_t^D = \mathcal{N}^D(w_t, q_t, \theta_t, k_{Dt}), \qquad N_t^O = \sum_{j \in \mathcal{J}} N_j(w_t, r_t, q_t, k_{Dt}; \alpha_j^R, \alpha_j^C). \tag{E.13}$$

Total employment and unemployment are given by equation (E.2). In this closure, data center expansion can increase direct employment while simul- taneously reducing employment in some industries through higher local factor prices and increasing employment in others through complementary demand or production linkages. Because wages cannot fall below $\bar{w}_t$, weak labor demand shows up as unemployment rather than lower wages.

E.5 Households

Households are indexed by $h \in \mathcal{H}$ and are heterogeneous in labor endow- ments, asset holdings, exposure to local prices, and housing positions. House- hold h chooses consumption c_{ht}, labor supply n_{ht}, and savings $b_{h,t+1}$ to max- imize expected discounted utility

$$U_h = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u_h(c_{ht}, 1 - n_{ht}, a_{ht}, \Omega(R_t)) \right], \tag{E.14}$$

where c_{ht} is consumption, $1 - n_{ht}$ is leisure, a_{ht} captures amenities or local quality of life, $\Omega(R_t)$ captures local stress from resource use, and $\beta \in (0, 1)$ is the discount factor.

The household faces a period budget constraint

$$c_{ht} + b_{h,t+1} = w_t n_{ht} + (1 + r_t) b_{ht}, \tag{E.15}$$

where w_t is the local wage, r_t is the local interest rate on savings, and b_{ht} is beginning-of-period financial assets.

Assumption 3 (Household heterogeneity and welfare exposure).

1. Period utility u_h is increasing in consumption and amenities and de- creasing in labor effort and stress $\Omega(R_t)$.

2. Households differ in wages, asset holdings, and exposure to stress (through $\Omega(R_t)$).

3. Household policy functions (choices of c_{ht}, n_{ht}, and $b_{h,t+1}$) are continu- ous in prices.

Households internalize the impact of resource use R_t through both utility and prices. Resource-intensive expansion affects utility directly via $u_h(\cdot, \Omega(R_t))$ and indirectly by shifting w_t, q_t, and possibly a_{ht}.

E.6 Welfare and the Government Problem

Per-period local welfare is defined as

$$\mathcal{W}(\theta_t, \sigma_t) = \sum_{h \in \mathcal{H}} \omega_h \, u_h(c_{ht}, 1 - n_{ht}, a_{ht}, \Omega(R_t)) + \sum_{j \in \mathcal{J}} \omega_j \, Y_{jt} + \omega_D y_{Dt} - \Omega(R_t), \tag{E.16}$$

where:

- ω_h are household welfare weights,
- Y_{jt} is output in non-data center industry j,
- y_{Dt} is output of the data center,
- ω_j and ω_D are weights on industry and data center production out- comes,
- $\Omega(R_t)$ is a reduced-form stress cost capturing congestion, reliability pressure, environmental stress, and unpriced local harms from heavy resource use.

The government chooses subsidy generosity σ_t to solve

$$W(\theta_t) = \max_{\sigma_t} \left\{ \mathcal{W}(\theta_t, \sigma_t) - C^{\text{fisc}}(\sigma_t) + \beta \, \mathbb{E}\left[W(\theta_{t+1}) \mid \theta_t, \sigma_t \right] \right\}, \tag{E.17}$$

where $C^{\text{fisc}}(\sigma_t)$ denotes the fiscal cost of subsidies, tax expenditures, and infrastructure spending associated with generosity σ_t. Since the demand state affects welfare through the price of data center output and the induced expansion of capacity, the planner's subsidy rule is shaped by expectations over future demand realizations rather than by any direct demand effect on physical technology.

Assumption 4 (Welfare and public finance).

1. $\Omega(R)$ is increasing and convex in resource use.

2. Fiscal costs $C^{\text{fisc}}(\sigma)$ are increasing and convex in subsidy generosity.

3. The planner attaches positive weight to household welfare and to local production outcomes, so that all weights $\{\omega_h\}_{h\in\mathcal{H}}$, $\{\omega_j\}_{j\in\mathcal{J}}$, and ω_D are nonnegative.

Under these assumptions, the government chooses a subsidy rule $\{\sigma_t\}$ based on the ex ante distribution of future demand states $\{\theta_t\}$ and the in- duced distribution of household welfare. Realized welfare is state contingent and can differ sharply from ex ante expectations in high-demand, high-scaling states, especially when resource-intensive industries are large and local stress costs are steep.

E.7 Employment Responses and Crowding Out

The key employment margin in the model is the response of total employment to a subsidy-induced or demand-induced expansion in data center activity. Let

$$\Delta N_t = \Delta N_t^D + \sum_{j\in\mathcal{J}} \Delta N_{jt}, \tag{E.18}$$

where ΔN_t^D is the direct change in data center employment and ΔN_{jt} is the change in employment in industry j.

For each industry j, a first-order approximation implies

$$\Delta N_{jt} \approx \frac{\partial N_j}{\partial k_{Dt}} \Delta k_{Dt} + \frac{\partial N_j}{\partial q_t} \Delta q_t. \tag{E.19}$$

The first term captures gains from complementarity with data center scale, while the second captures losses from higher local resource prices.

Lemma 1 (Gradual expansion)**.** *Under Assumption 1, an increase in subsidy generosity σ_t or in the demand state θ_t weakly raises the shadow value of installed data center capital and therefore weakly raises optimal investment i_{Dt}.*

Sketch. A more generous subsidy raises the payoff associated with installed capital through $S_D(\sigma_t, k_{Dt}, i_{Dt})$. A higher demand state raises the output price $p^D(\theta_t)$ and therefore the marginal revenue from a given level of output. By concavity of production and convexity of Γ_D, the first-order condition for investment equates the marginal value of additional capital to the marginal purchase and adjustment cost. Since marginal adjustment cost is increasing in i_{Dt}, positive shocks to σ_t or θ_t increase investment but do not induce an infinite jump in k_{Dt}. Hence expansion is monotone and gradual. □

Lemma 2 (Stress transmission)**.** *If data center investment raises k_{Dt} and resource use r_{Dt}, then the stress cost Ω_t weakly rises.*

Sketch. By equation (E.11), an increase in data center scale raises total resource use R_t through higher r_{Dt}. Since equation (E.12) implies that Ω is increasing in R_t, higher R_t weakly raises Ω_t. □

Proposition 1 (Industry-level employment response)**.** *Let industry j be characterized by resource intensity α_j^R and complementarity α_j^C. Following a subsidy or demand shock that raises data center scale ($\Delta k_{Dt} > 0$) and lo- cal resource prices ($\Delta q_t > 0$), industry-j employment satisfies equation* (E.19)*. Then:*

1. *$\Delta N_{jt} > 0$ is more likely when complementarity α_j^C is large, resource intensity α_j^R is small, and Δq_t is modest.*

2. *$\Delta N_{jt} \leq 0$ is more likely when resource intensity α_j^R is large, comple- mentarity α_j^C is small, and Δq_t is large.*

Sketch. By Assumption 2, the positive term $\frac{\partial N_j}{\partial k_{Dt}} \Delta k_{Dt}$ is larger for industries with higher complementarity α_j^C, while the negative term $\frac{\partial N_j}{\partial q_t} \Delta q_t$ is more negative for industries with higher resource intensity α_j^R. The sign of ΔN_{jt} therefore depends on whether the complementary gain from higher data cen- ter scale dominates the price-induced employment loss. □

Proposition 2 (Net employment gains versus crowding out)**.** *Suppose a sub- sidy or demand shock raises data center activity so that* $\Delta N_t^D > 0$ *and induces changes in non-data center industry employment according to equation* (E.19)*. Then:*

1. *Net local employment grows if and only if*

$$\Delta N_t^D + \sum_{j \in \mathcal{J}} \Delta N_{jt} > 0.$$

2. *Net local employment does not grow if and only if*

$$\Delta N_t^D + \sum_{j \in \mathcal{J}} \Delta N_{jt} \leq 0.$$

Equivalently, crowding out dominates whenever

$$\left| \sum_{j \in \mathcal{J}} \Delta N_{jt} \right| \geq \Delta N_t^D \qquad \textit{when} \qquad \sum_{j \in \mathcal{J}} \Delta N_{jt} < 0. \tag{E.20}$$

Sketch. Equation (E.18) gives the decomposition of aggregate employment into direct data center employment and the sum of industry-level employment changes outside the data center sector. Net employment rises if the direct employment gain and the employment gains in complementary industries jointly exceed the losses in resource-intensive industries. It does not rise when these losses weakly dominate. □

Proposition 3 (Conditions for net employment growth)**.** *Holding fixed the path of demand shocks* $\{\theta_t\}$*, net local employment is more likely to grow after a subsidy or favorable demand shift when:*

1. *data center production is relatively labor intensive, so that* $\dfrac{\partial N_t^D}{\partial \theta_t}$ *and* $\dfrac{\partial N_t^D}{\partial \sigma_t}$ *are large;*
2. *local resource supply is relatively elastic, so that* $\dfrac{\partial q_t}{\partial R_t}$ *is small;*

3. *the local industry composition places greater weight on industries with high complementarity α_j^C and lower weight on industries with high re- source intensity α_j^R;*

4. *the local labor market begins with slack, so that wage pressures from expansion are limited.*

Conversely, net employment is less likely to grow when data centers are highly capital and resource intensive, local resource supply is inelastic, and the local economy contains many industries that are strongly exposed to local resource prices but weakly complementary to data center activity.

Sketch. The direct employment gain ΔN_t^D is larger when data centers are more labor intensive and when subsidy generosity or stronger demand trans- lates strongly into scale. The aggregate indirect employment response $\sum_j \Delta N_{jt}$ is less negative, or more positive, when the price response Δq_t to resource use is muted and when a larger share of industries has high α_j^C rather than high α_j^R. Slack labor markets additionally limit wage-driven crowding out in non-data center industries. □

Proposition 4 (Ex ante support and ex post regret)**.** *Suppose policymakers evaluate a project using expected welfare before the realization of θ_t, while realized welfare is state contingent. If low and moderate realizations of θ_t generate manageable stress costs but high realizations generate sharply in- creasing stress through the convex function $\Omega(R)$, then there exist parameter values such that expected ex ante welfare from subsidizing the project is pos- itive while ex post welfare is negative in high-demand states.*

Sketch. Consider the period welfare expression in equation (E.16). Benefits from expansion rise with the demand-induced value of data center output, direct employment, and employment gains in complementary industries, but local costs enter through $\Omega(R_t)$, which is convex in R_t. Convexity implies that as the high-demand state pushes resource use far enough upward, marginal and total stress costs can increase more than proportionally.

One can choose a distribution for θ_t such that expected welfare, averaging across states, re- mains positive, yet realized welfare in sufficiently high states becomes nega- tive. □

E.8 Testable Implications

The model yields a set of empirical predictions that map naturally to industry- level heterogeneity.

- **Direct versus net jobs.** Announcements, subsidies, or openings should increase employment within the data center sector, but total local employment rises only if the direct gain plus the gains in comple- mentary industries exceed the losses in resource-intensive industries.
- **Resource intensity heterogeneity.** Industries with higher measured resource intensity should experience weaker employment growth, or outright employment declines, after data center entry or subsidy adop- tion.
- **Complementarity heterogeneity.** Industries with higher measured complementarity to data centers should experience stronger employ- ment growth after data center expansion.
- **Industry mix as a moderator.** Places with an industry mix tilted to- ward high-α_j^R industries should exhibit more crowding out and smaller net employment gains, while places tilted toward high-α_j^C industries should exhibit stronger net employment growth.
- **Demand-state heterogeneity.** Locations exposed to larger demand- driven expansions in data center scale should experience larger direct employment gains but also stronger pressure on local resources and therefore greater scope for crowding out.
- **Stronger effects where resource supply is less elastic.** Net em- ployment, wage, and price effects should be larger in places where local resource prices respond more sharply to increases in aggregate resource use.